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Introduction

This project presents the design and development of MediCare Plus, a web-based medical management system created to streamline communication and administrative tasks between patients, doctors, and administrators. The primary objective of the system is to digitalize essential hospital functions such as appointment scheduling, patient record management, test report handling, prescription access, secure messaging, and feedback collection.

The system was developed using HTML, CSS, PHP, MySQL, and JavaScript, ensuring a functional, user-friendly, and secure platform. Three major user roles;Admin, Doctor, and Patient, were implemented with separate dashboards to maintain data privacy through role-based authentication.

The project followed structured software engineering principles including requirements analysis, interface design, wireframing, database modeling, and system testing. Special attention was given to usability, consistency, data validation, session security, and responsive UI design. The final solution aims to provide a seamless digital experience that reduces manual workload and enhances the overall healthcare service quality.

Task 01

a) Website Comparison

1. Asiri Health

- Asiri Health, Sri Lanka's largest private hospital network. Services offered include online bookings, lab test result access, health-check packages, and an international patient desk.
- Website: <https://www.asirihealth.com>

2. Hemas Hospitals

- Hemas Hospitals stands out as a prominent private healthcare provider in Sri Lanka, recognized for its dedication to high-quality care
- Website: <https://www.hemashospitals.com>

3. Durdans Hospital

- A well-established private healthcare facility located in Colombo, providing cutting-edge diagnostic services, surgical procedures, and specialized care.
- Website: <https://www.durdans.com>

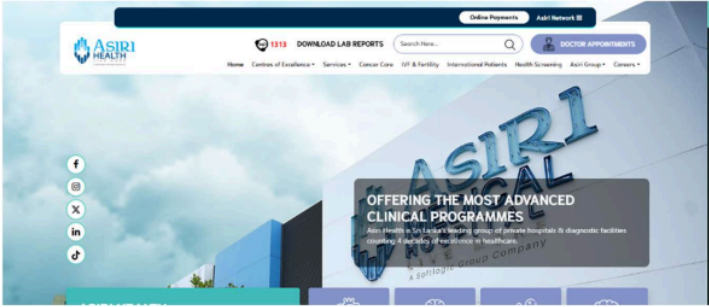
4. Apollo Hospitals

- Apollo Hospitals is recognized as one of the largest healthcare groups in Asia. The robust branding and extensive digital integration exemplify the effectiveness of technology-driven healthcare delivery.
- Website: <https://www.apollohospitals.com>

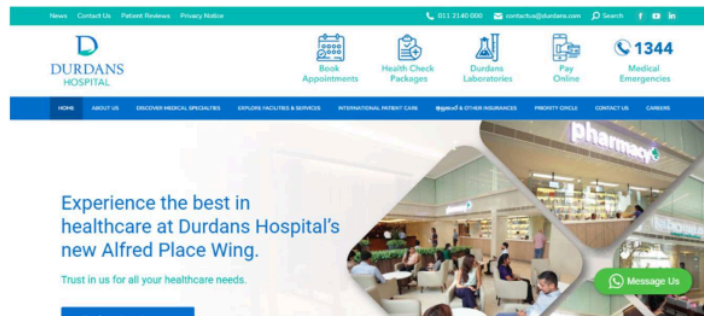
5. Mayo Clinic

- The Mayo Clinic is internationally esteemed for its outstanding contributions to medical research and patient care.
- Website: <https://www.mayoclinic.org>

1. Home Page Comparison

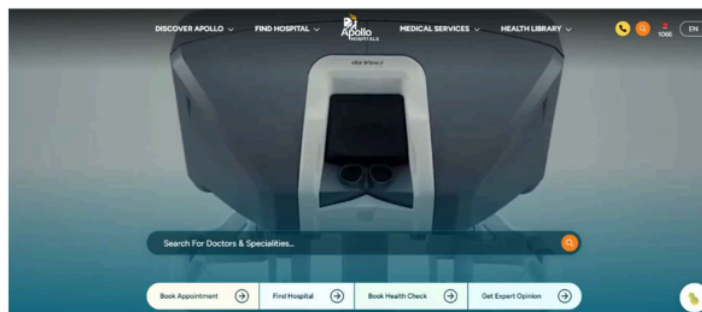
Hospital	
Asiri Hospital	 • Their home page features a large hero section that showcases their branding. • They quickly guide users to the “Centers of Excellence,” available services, and hospital locations.
Hemas Hospitals	 • Their homepage highlights that they are "internationally accredited, multi-specialty family hospitals" located in Sri Lanka. • They provide information about their locations. The layout is simpler compared to others, focusing on brand trust.

Durdans
Hospital



- The home page emphasizes “Best private Hospital in Sri Lanka”.
- They also highlight international patient care.
- So the landing page supports both an overseas audience.

Apollo
Hospitals



- The home page is titled “We Care for You, Every Step of the Way!” and emphasizes international standards and reach.
- The main page likely has strong global brand pull.
- They may direct quickly to contact.

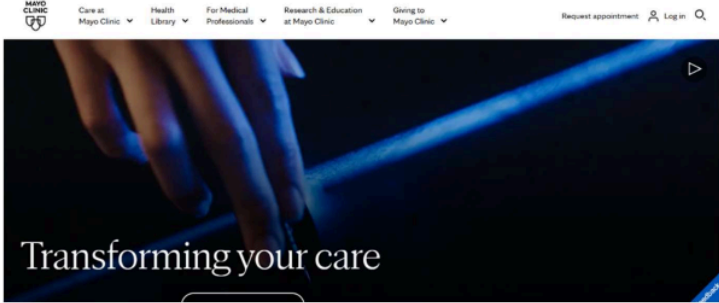
Mayo Clinic	 • The home page emphasized “Top-ranked hospital”. • Mayo is a major international brand, the landing likely focuses on credibility, breath of care, and allows deep navigation, • More complex design
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Table 1

2. Navigation Bar

Hospital	
Asiri Hospital	 • Fixed, responsive bar with dropdowns for Doctors, Services, Locations, and Contact Us. • Includes quick access to emergency numbers and hospital branches.
Hemas Hospitals	 • Simple, family-friendly homepage highlighting the hospital's accreditation and modern facilities. • Uses real patient imagery and welcoming text to build emotional connection.
Durdans Hospital	 • Well-structured menu with Find a Doctor, Book Appointment, Health Packages, and Patient Care. • Dropdowns are well organized, improving navigation speed.

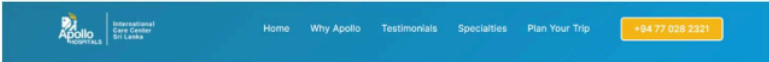
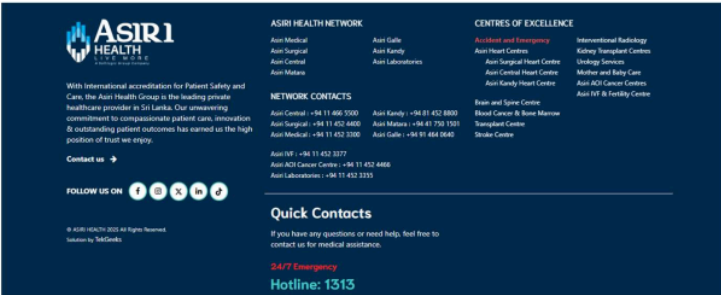
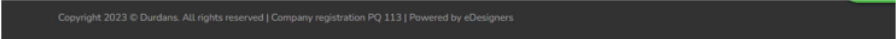
Apollo Hospitals	 • Comprehensive menus for Specialties, Patients, Doctors, etc. • Dropdown menus are organized by departments for better usability.
Mayo Clinic	 • Detailed, multi-level navigation with tabs for Patients & Visitors, Research, Education, and Departments. • Search bar integrated for direct access to medical information.

Table 2

3. Footer

Hospital	
Asiri Hospital	 • Contains contact details and quick links. • Displays copyright and social media icons.
Hemas Hospitals	 • Displays hotline numbers, inquiry links, and branch contact details. • Social media links encourage user engagement.
Durdans Hospital	 • Includes location maps, emergency numbers, and a feedback form. • Contact information presented in both local and international formats.

Apollo Hospitals	 The screenshot shows the footer of the Apollo Hospitals website. It includes the Apollo Hospitals logo, a brief statement about the group's impact, social media icons, and a grid of links categorized under SERVICES, APOLLO INDIA, RESOURCES, and SUPPORT. The footer also contains copyright information for 2023. • Displays global branch details, privacy policy, and patient support links. • Includes newsletter sign-up and accreditation logos.
Mayo Clinic	 The screenshot shows the footer of the Mayo Clinic website. It features a light blue background with a grid of links for Terms & Conditions, Privacy Policy, Notice of Privacy Practices, Notice of Nondiscrimination, Digital Accessibility Statement, Advertising & Sponsorship Policy, and Site Map. Below this is copyright information for 1998-2025 and a language selector set to English. • Extensive footer with the links to About us, Careers, Contact, Research, and Privacy Policy. • Includes global partner links and social-media platforms.

Table 3

4. Color Theme

Hospital	
Asiri Hospital	• Blue and White • Clean and professional • Highlights trust and medical purity.
Hemas Hospitals	• Light Blue and green tones create a calm and healing atmosphere. • White background ensures content clarity.
Durdans Hospital	• Blue-gray palette promotes modern elegance and calmness. • Icons and buttons are consistent across all pages.
Apollo Hospitals	• Deep blue with golden branch details, privacy policy, and patient support links. • High contrast fonts improve visibility.
Mayo Clinic	• Blue and white combination for professionalism and credibility. • Clean typography enhances readability.

Table 4

5. User Experience

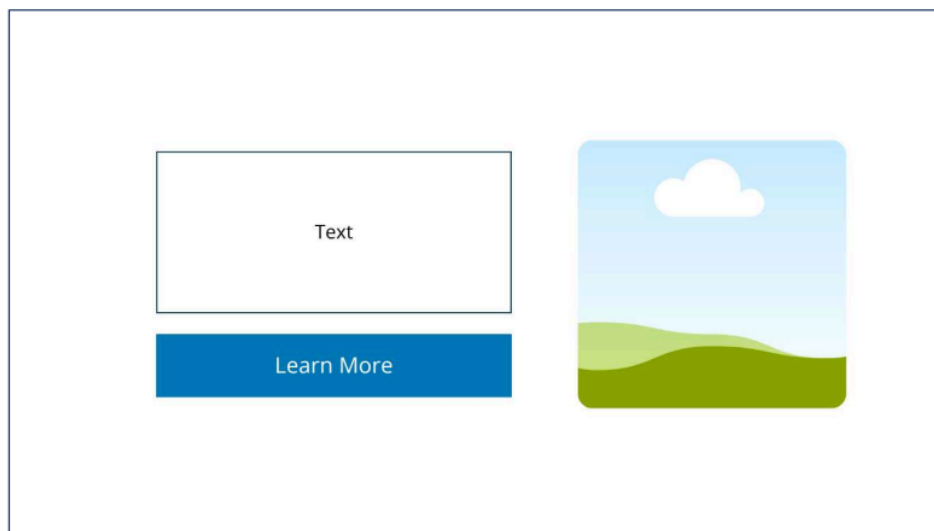
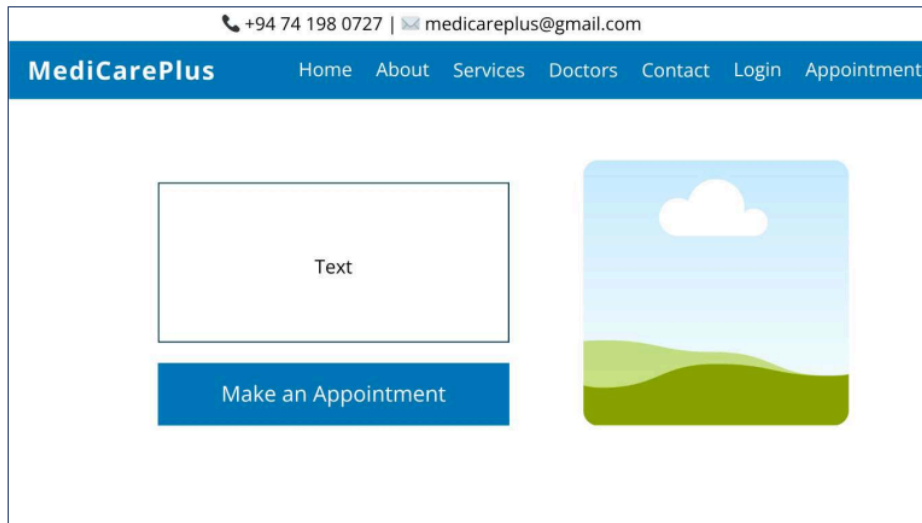
Hospital	
Asiri Hospital	• Intuitive layout with minimal scrolling and quick page loading. • Easy to navigate. • Patients can reach booking pages within two clicks.
Hemas Hospitals	• Mobile responsive design with simple flow between pages. • Provides quick online booking and health check access.
Durdans Hospital	• Smooth animations and clear hierarchy improve readability. • Booking and inquiry options are easily accessible.
Apollo Hospitals	• Very responsive, fast, and professional look. • Maintains visual consistency and quick access to vital information.
Mayo Clinic	• Highly responsive, structured for global accessibility. • Maintains excellent organization even with large content volumes.

Table 5

b) Diagrams and site maps

1. Wireframe Diagrams

1. Home Page Wireframe



Heading



Text



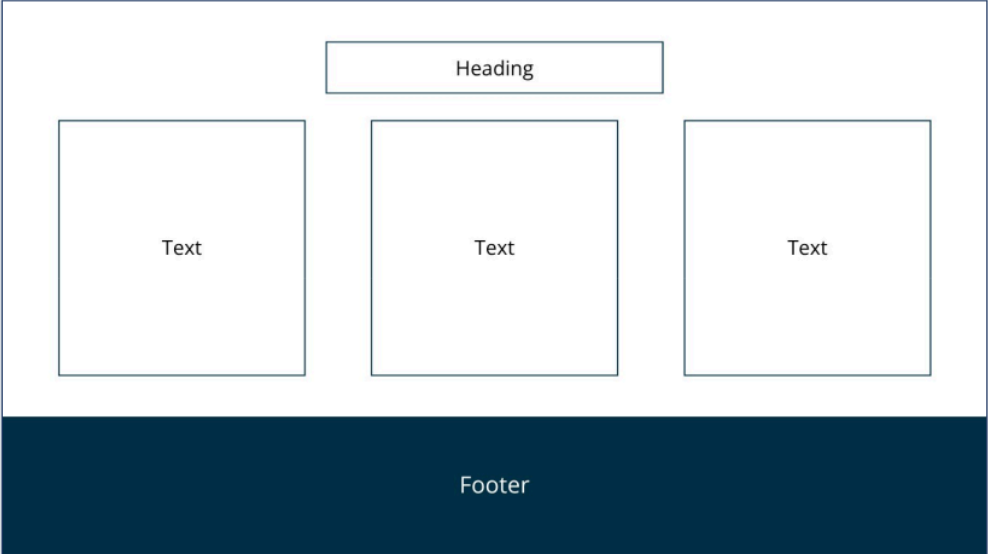
Text



Text

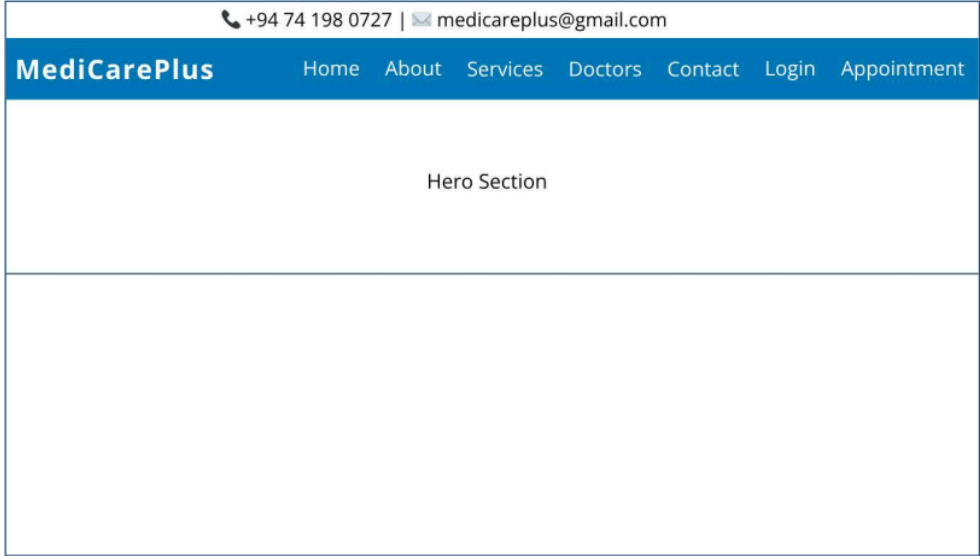
Heading

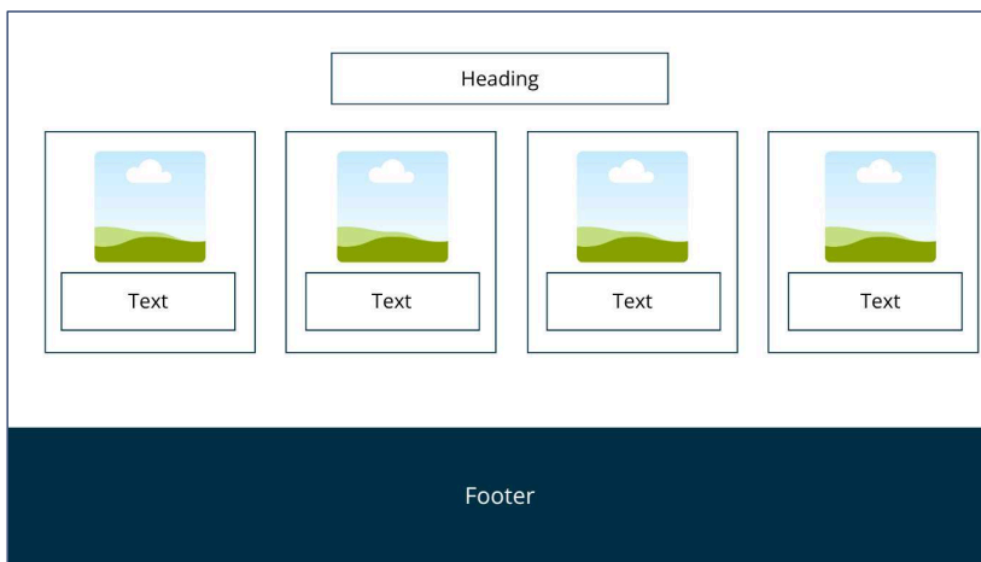
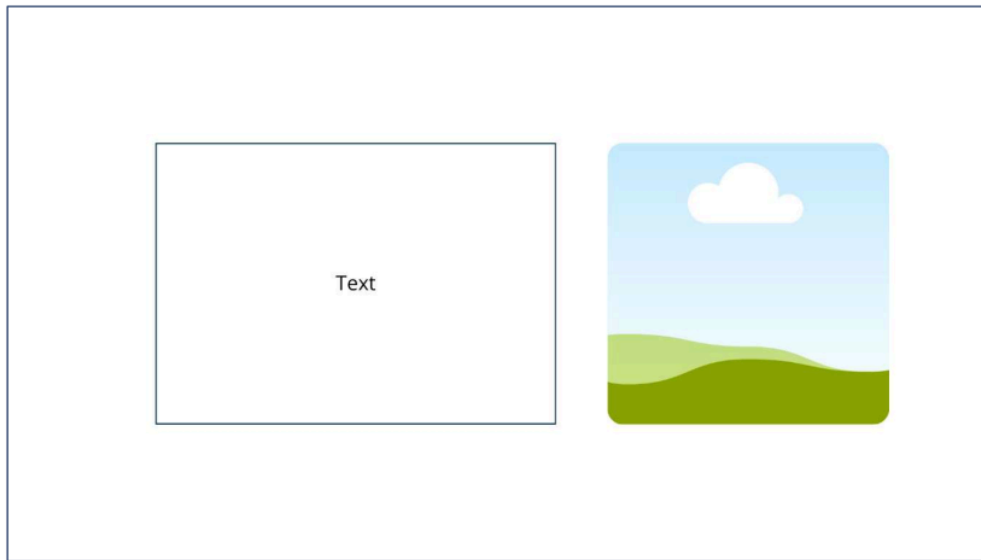
Text



This wireframe offers a clear first impression of the system, highlighting main navigation elements (Home, Services, Doctors, Contact, Login) in a simple layout. It emphasizes usability and easy access to essential information.

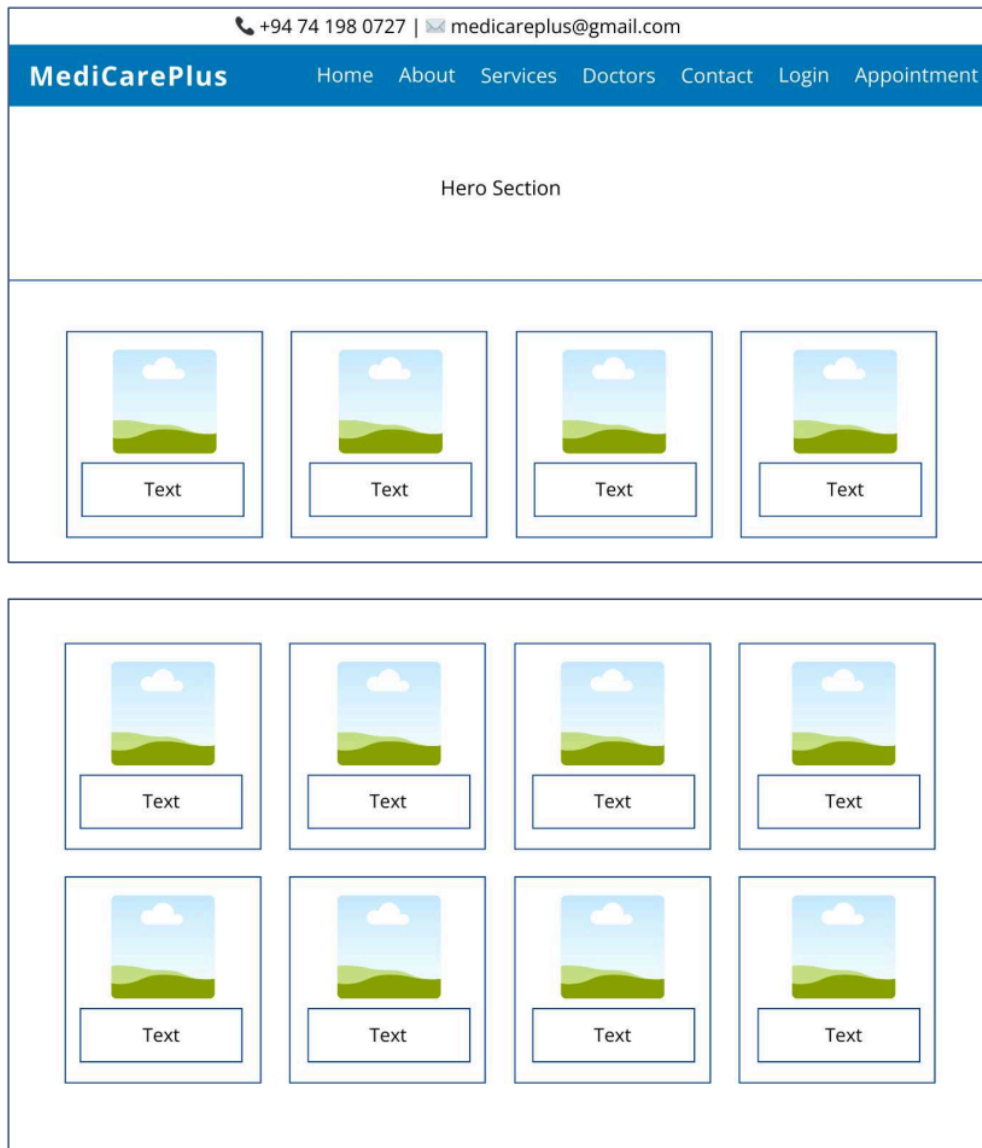
2. About Page Wireframe

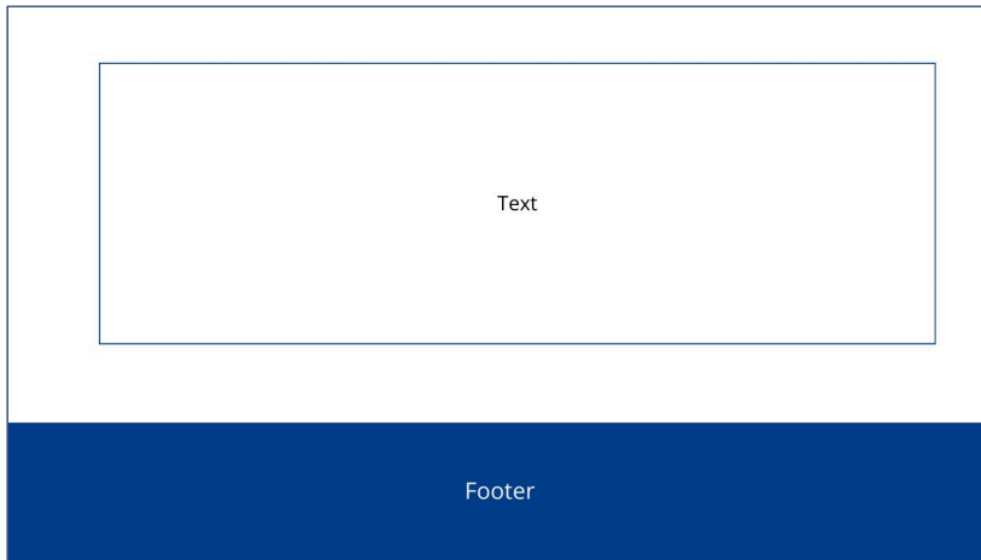




The About page wireframe outlines the system's purpose and the organization behind it. With a clean and informative layout, it presents the background, mission, and core values to foster trust and transparency for new users.

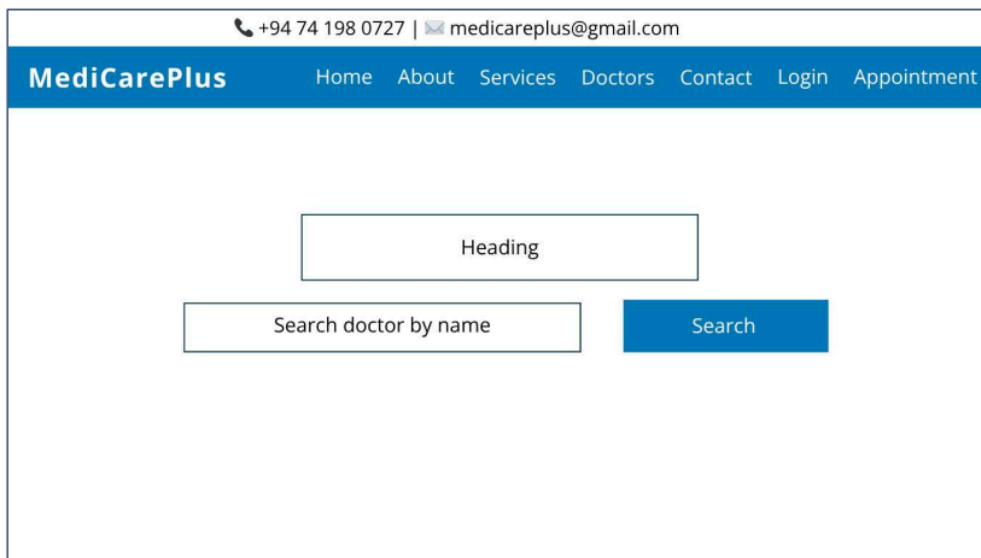
3. Services Page Wireframe





This wireframe presents medical services in a structured grid, allowing patients to quickly browse and categorize options for easy decision-making.

4. Doctors Page Wireframe



Doctor	Specialization	Email	Phone

Footer

The Doctors page wireframe uses a card format to display doctors' names, specializations, and contact details, facilitating easy comparison for patients. The layout supports filtering and enhances user engagement for seamless doctor selection.

5. Contact Page Wireframe

+94 74 198 0727 | medicareplus@gmail.com

MediCarePlus

HomeAboutServicesDoctorsContactLoginAppointment

Heading

Text

Send a Message

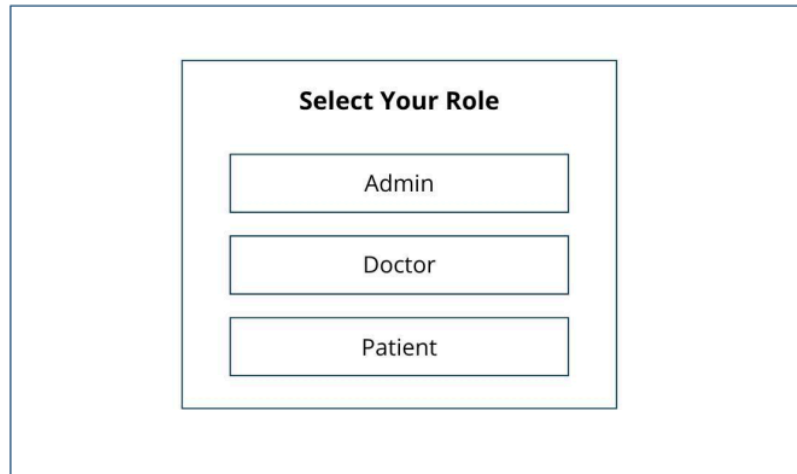
Form

Send Message

Footer

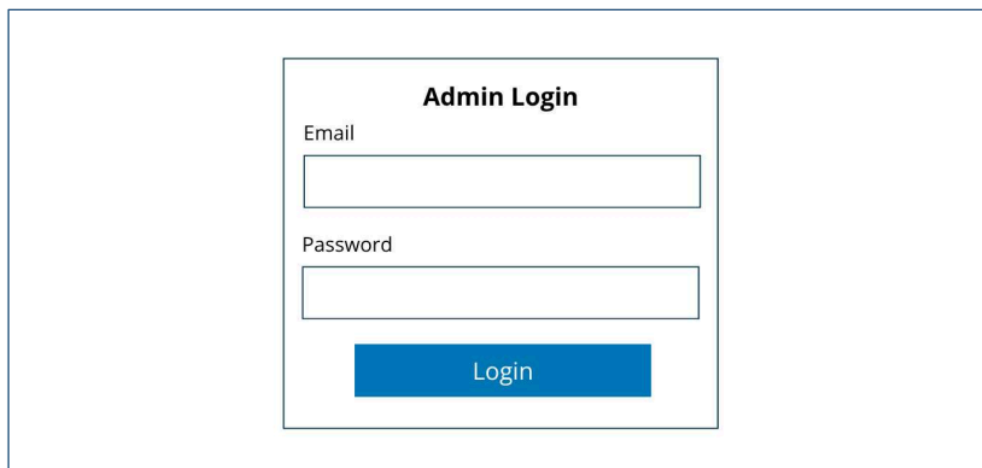
The Contact Page wireframe features a simple form and clear contact details for quick communication with support. The tidy design makes form-filling easy.

6. Role Section Wireframe



A wireframe for a role selection interface. It consists of a large outer rectangle containing a smaller inner rectangle. The inner rectangle has a title "Select Your Role" at the top. Below the title are three stacked rectangular buttons, each containing the text "Admin", "Doctor", and "Patient" respectively.

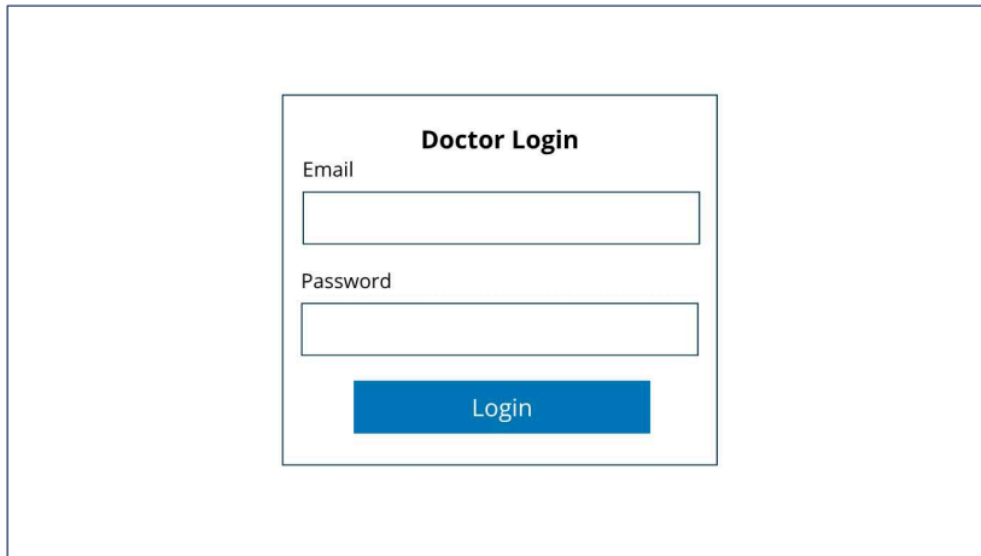
7. Admin Login Wireframe



A wireframe for an admin login interface. It consists of a large outer rectangle containing a smaller inner rectangle. The inner rectangle has a title "Admin Login" at the top. Below the title are two input fields: the first is labeled "Email" and the second is labeled "Password". Below the input fields is a blue button with the text "Login".

This wireframe offers a simple and user-friendly login form for admin.

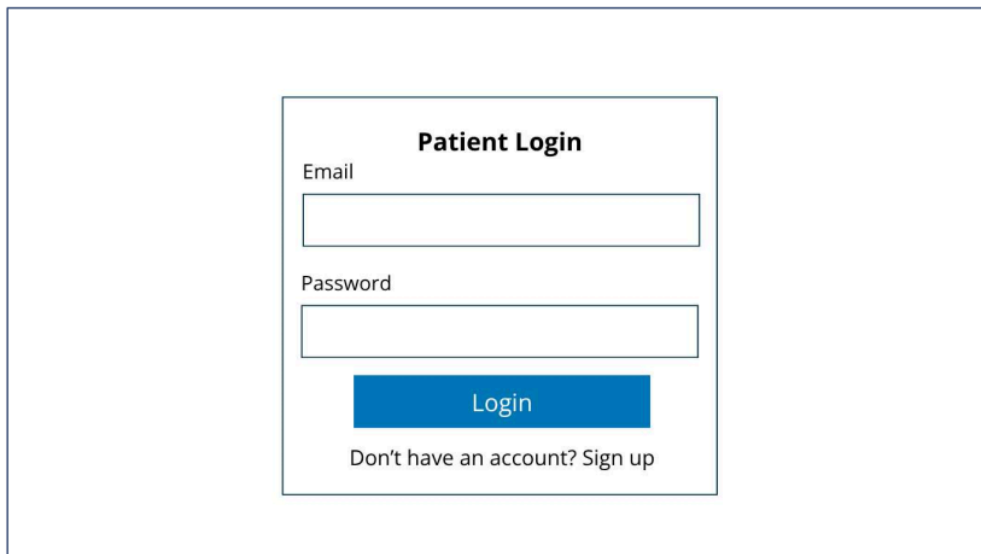
8. Doctor Login Wireframe



The wireframe for the Doctor Login form is contained within a large rectangular frame. Inside this frame is a smaller box representing the login form. At the top of this box is the title "Doctor Login" in bold. Below the title are two input fields: the first is labeled "Email" and the second is labeled "Password". Both labels are positioned to the left of their respective input boxes. At the bottom of the form box is a blue rectangular button with the text "Login" in white.

This wireframe ¹²offers a simple and user-friendly login form for doctor.

9. Patient Login Wireframe



The wireframe for the Patient Login form is contained within a large rectangular frame. Inside this frame is a smaller box representing the login form. At the top of this box is the title "Patient Login" in bold. Below the title are two input fields: the first is labeled "Email" and the second is labeled "Password". Both labels are positioned to the left of their respective input boxes. At the bottom of the form box is a blue rectangular button with the text "Login" in white. Below the button is a link that says "Don't have an account? Sign up".

This wireframe offers a simple and user-friendly login form for patients.

10. Patient Signup Wireframe

Create Patient Account

Full name

Email

Phone

Password

Create Account

Already have an account? Login

This wireframe organizes essential patient details section-by-section to reduce errors and streamline onboarding, enhancing user confidence when submitting sensitive information.

11. Admin Dashboard Wireframe

MediCarePlus

Welcome, System Admin

Dashboard

Manage Doctors

Manage Patients

Appointments

Payments

Reports

Logout

Admin Dashboard

Date

Text

Text

Text

Text

Quick Actions

Manage Doctors

Manage Patients

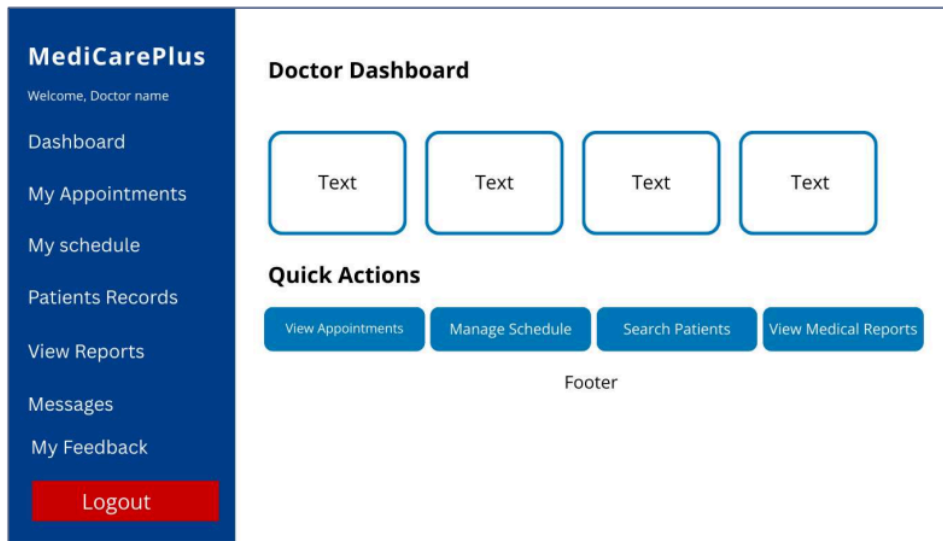
View Appointments

View Payments

Footer

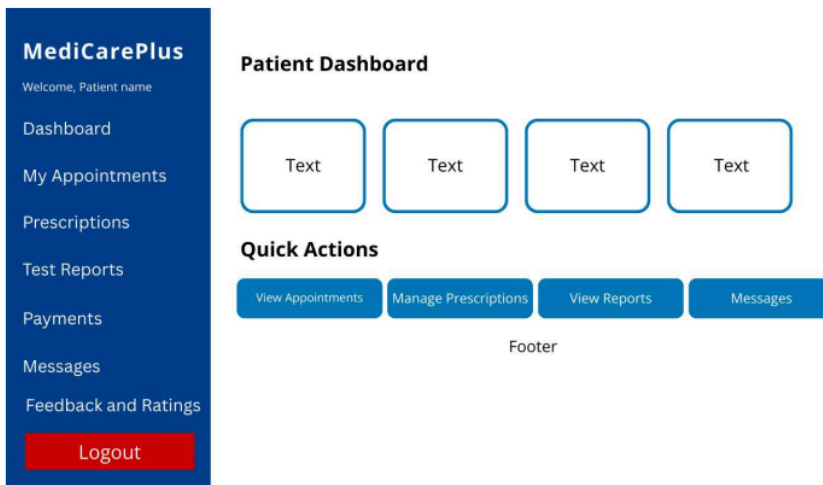
The wireframe uses cards and a sidebar for quick admin access and system metrics overview, enhancing workflow.

12. Doctor Dashboard Wireframe



This wireframe prioritizes daily tasks such as appointments, patients, messages, and schedules, providing doctors with immediate access to information in a clean, professional interface.

13. Patient Dashboard Wireframe



The patient dashboard wireframe provides direct access to appointments, prescriptions, reports, and payments. It ensures easy navigation and improves user experience by keeping everything grouped logically. Wireframes provided the structural foundation, while UI pages transformed those structures into a polished, interactive interface that improves the system's usability.

2. UI Design

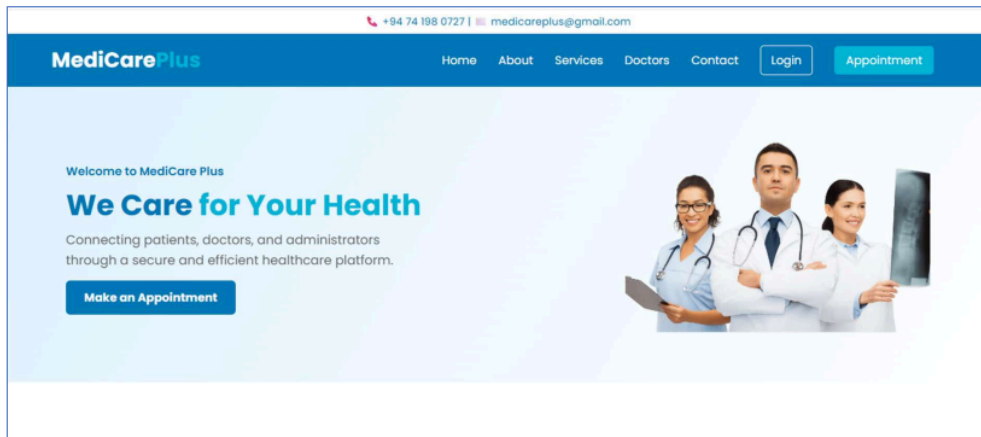


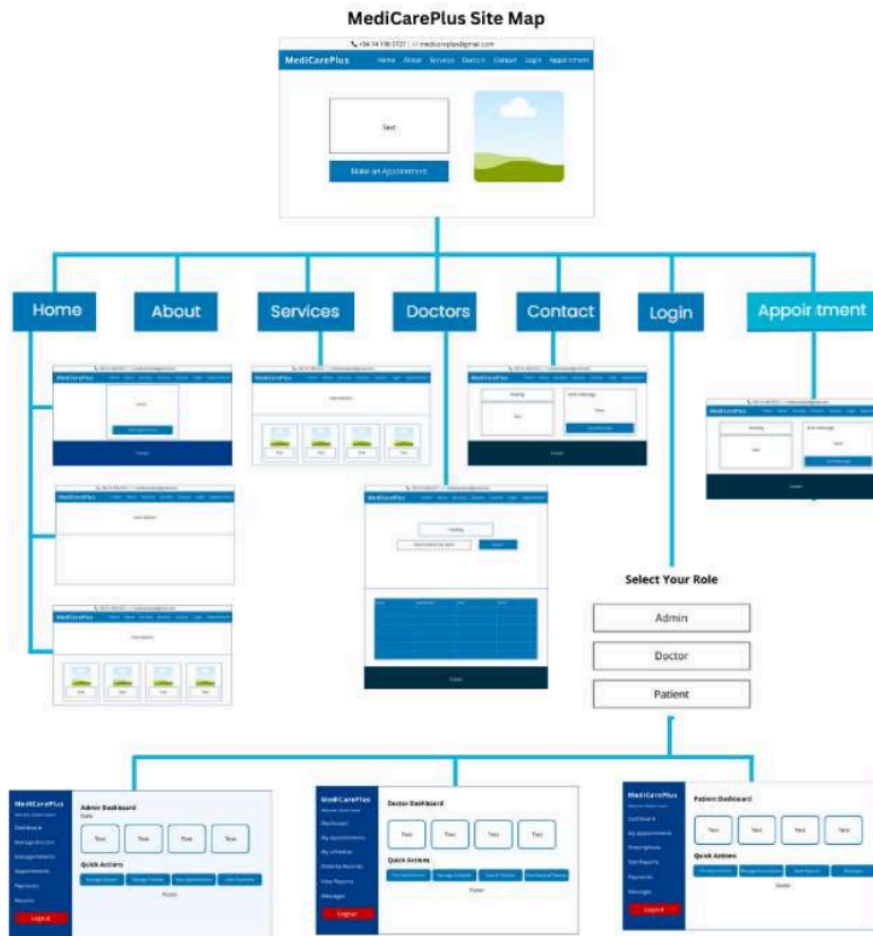
Figure 1

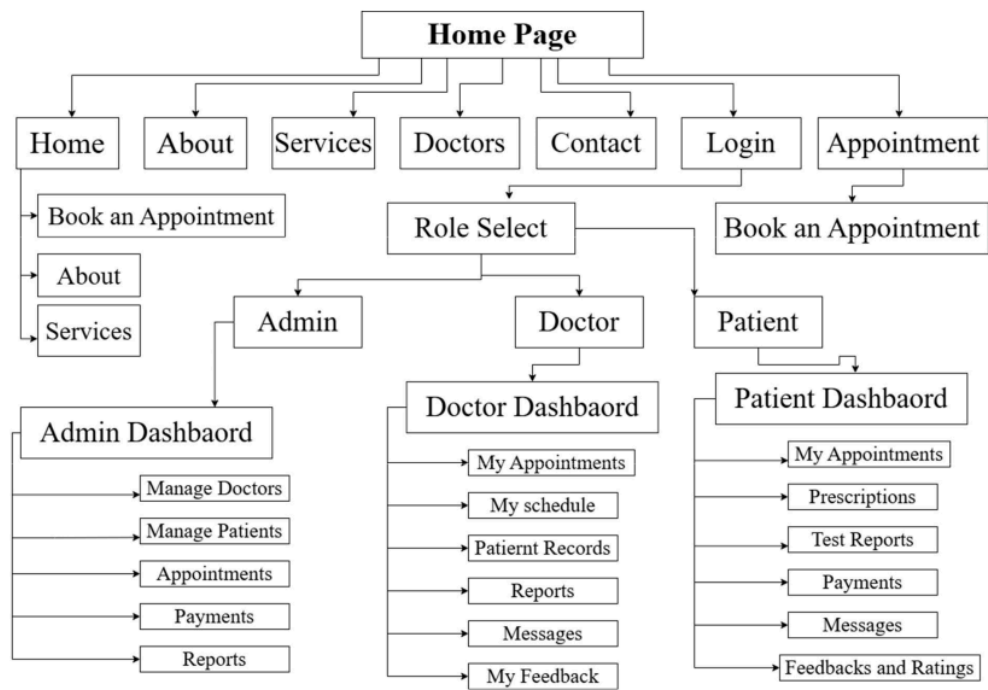
The user interface for MediCare Plus was designed following a modern, medical-friendly theme used on clean blues and minimalistic layouts. The focus is on accessibility, clarity, and ease of navigation.

- The final UI pages were designed with a blue medical-themed color palette (#0077b6, #00b4d8) to maintain professionalism.
- All pages follow consistent styling—cards, tables, action buttons, and sidebars—ensuring visual coherence.
- Pages were made responsive with flexible grid layouts and simplified navigation.
- Icons, spacing, and typography were improved from the wireframes to enhance readability and user interaction.

3. Sitemaps

Low-fidelity wireframes were created to outline the layout structure, key components, and flow. Sections include Hero, About, and Services: Doctors Directory, Contact, and Appointment modules.





The sitemap illustrates the MediCare Plus system's structure and navigation, showing main modules Admin, Doctor, Patient, and their sub-pages. It helps users, developers, and evaluators understand the hierarchy, features, and pathways, supporting intuitive navigation and consistency. Functional requirements like appointments, records, reports, payments, feedback, and messaging are logically grouped, making the system user-friendly and well-structured.

Task 02

a) Implementation

1. Frontend Implementation

The frontend is implemented using HTML, CSS, and JavaScript, following responsive and accessibility first principles. The Ui is modular, consistent across pages, and uses the Poppins font and a calm blue palette to convey trust and clarity.

- Top bar
- Hero Section
- Service cards and doctor cards
- Doctors' directory
- Role section popup
- Login/Signup
- Dashboards (Admin/Doctor/Patient)
- Appointment flow
- Contact form and feedback form

Technology	Purpose
HTML	Semantic layout
	Accessibility
CSS	Styling
	Responsive design
Flexbox and grids	To create responsive layouts
JS	Form validation
	UI interactivity

2. UI Features Implemented

Responsive Layout	• The website automatically adjusts for mobile and desktop. • Navigation bar collapses gracefully. • Tables adjust into scrollable containers.
Consistent Theme	• A full medical UI theme was applied • Primary color: #0077b6 • Secondary: #f0f9ff • Accent: #219ebc • Typography: Poppins for clean readability
Accessibility Enhancements	• High contrast for readability. • Buttons with focus/hover states. • Descriptive alt text for icons/images.
Reusable Components	• A reusable login card design. • Common dashboard layout for admin/doctor/patient. • Shared top bar/footer.

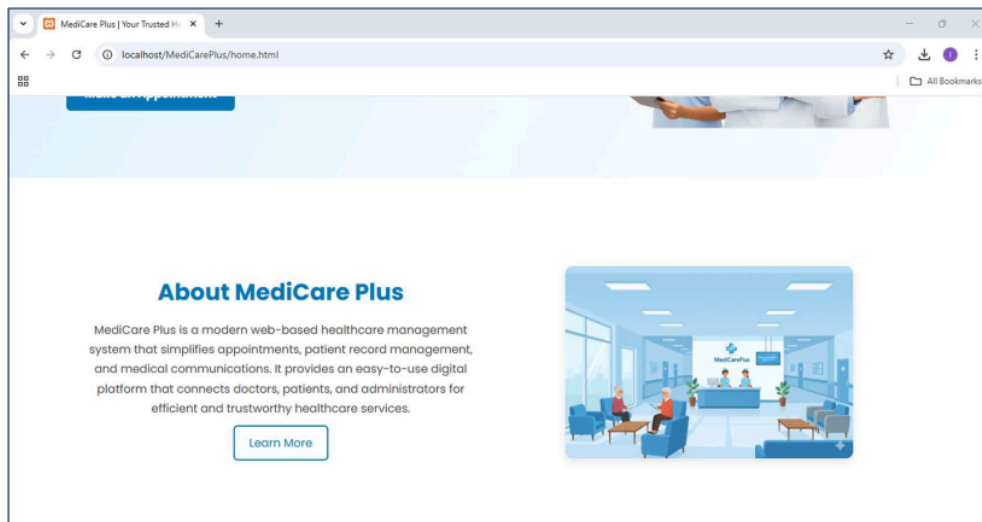
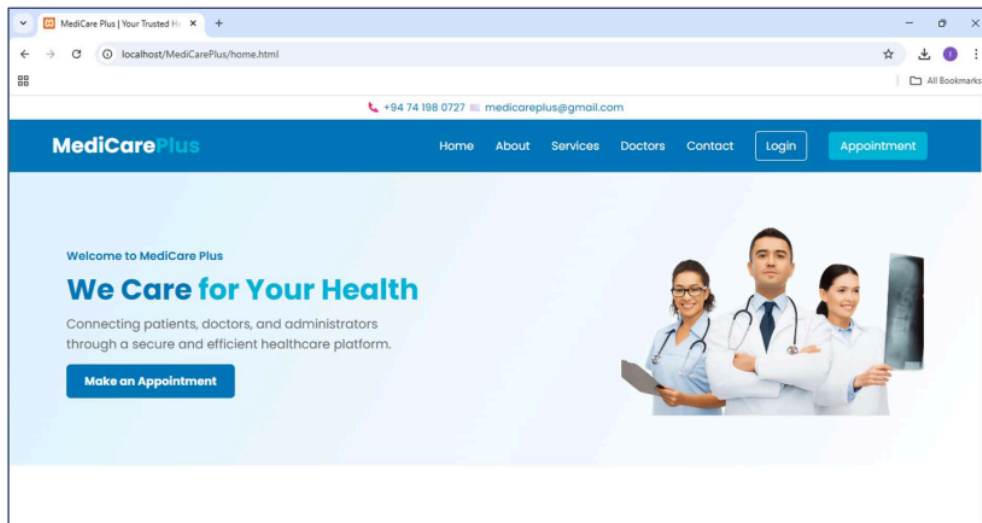
3. UI Design Justification

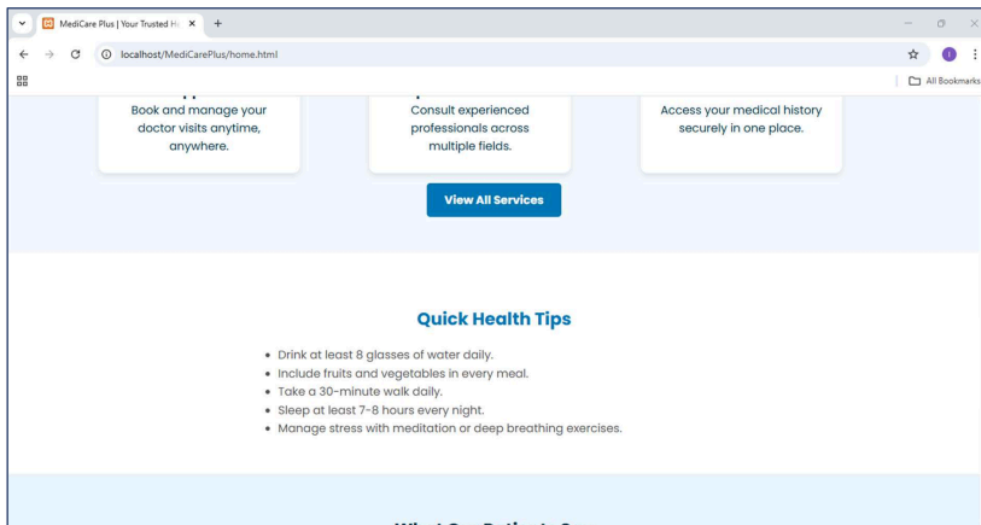
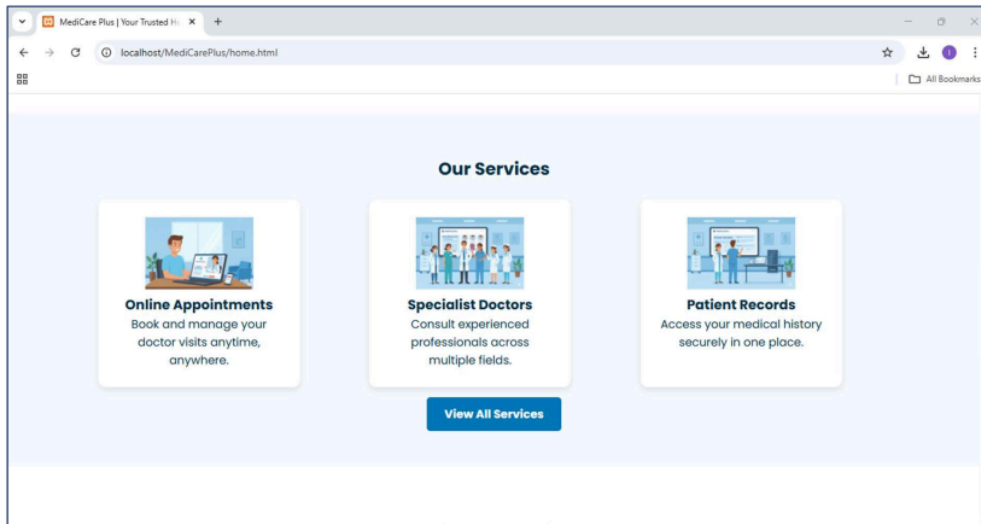
Blue Medical theme	Blue is psychologically linked to trust, cleanliness, and healthcare.
Rounded cards & shadows	Creates a friendly, modern visual hierarchy.
Large buttons	Improves accessibility and usability, especially on mobile.
Centered login form	Reduces cognitive load 'user knows exactly where to interact.
Sidebar dashboard design	Doctors/Admins access multiple tools → sidebar gives faster workflow.
Icons and text tables	Improves recognition speed and assists users with low literacy.
Minimalistic layout	Reduces distractions and ensures focus on main tasks.

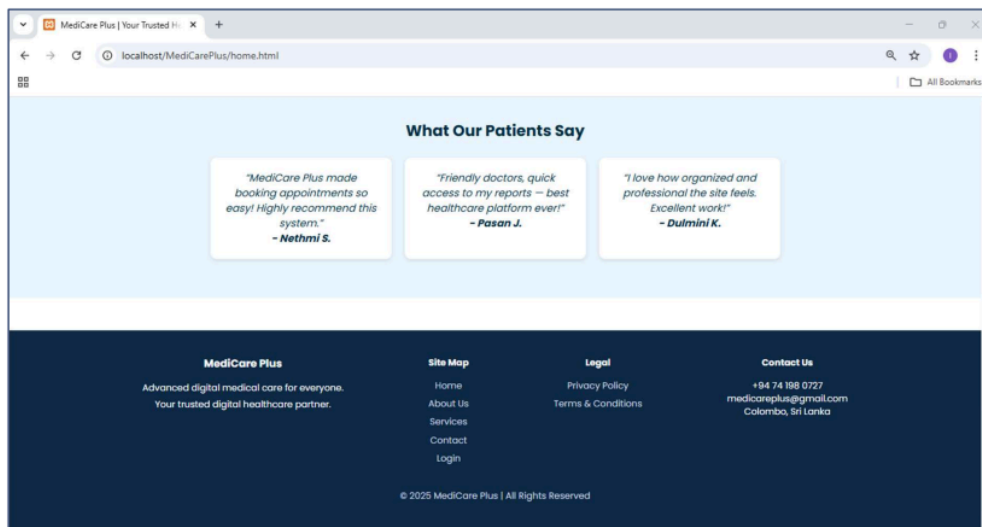
4. Implementation of UI Screens

User Pages

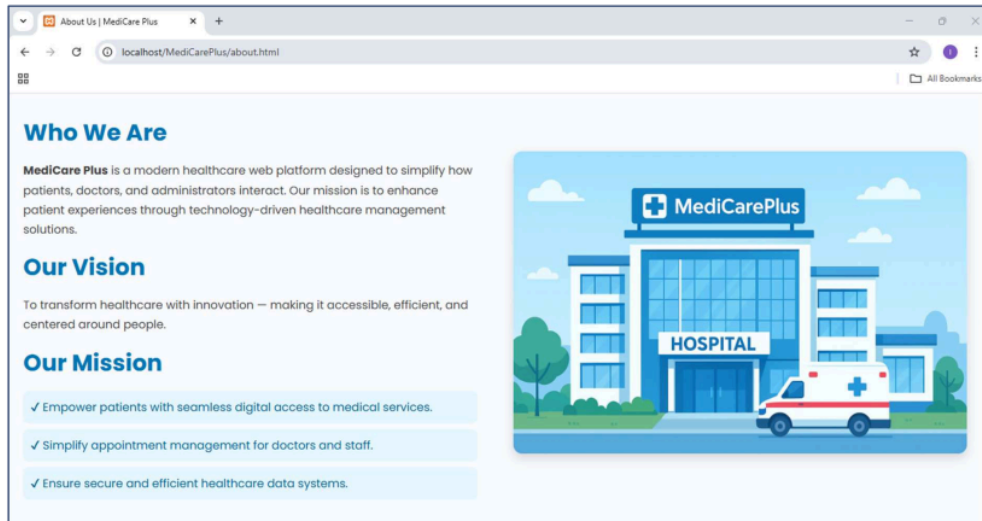
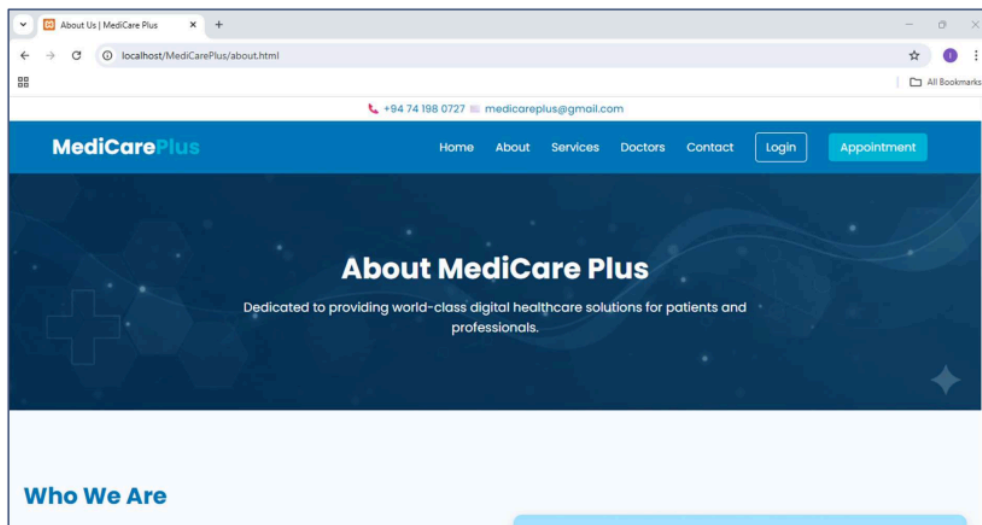
These public-facing pages are designed to be welcoming and informative. The home page introduces MediCare Plus and guides users to the login or registration sections. The About page explains the purpose of the system, while the Services page outlines medical services available. The Contact page allows users to reach out for assistance. Typography, spacing, and icons are used to maintain readability and professionalism. Home

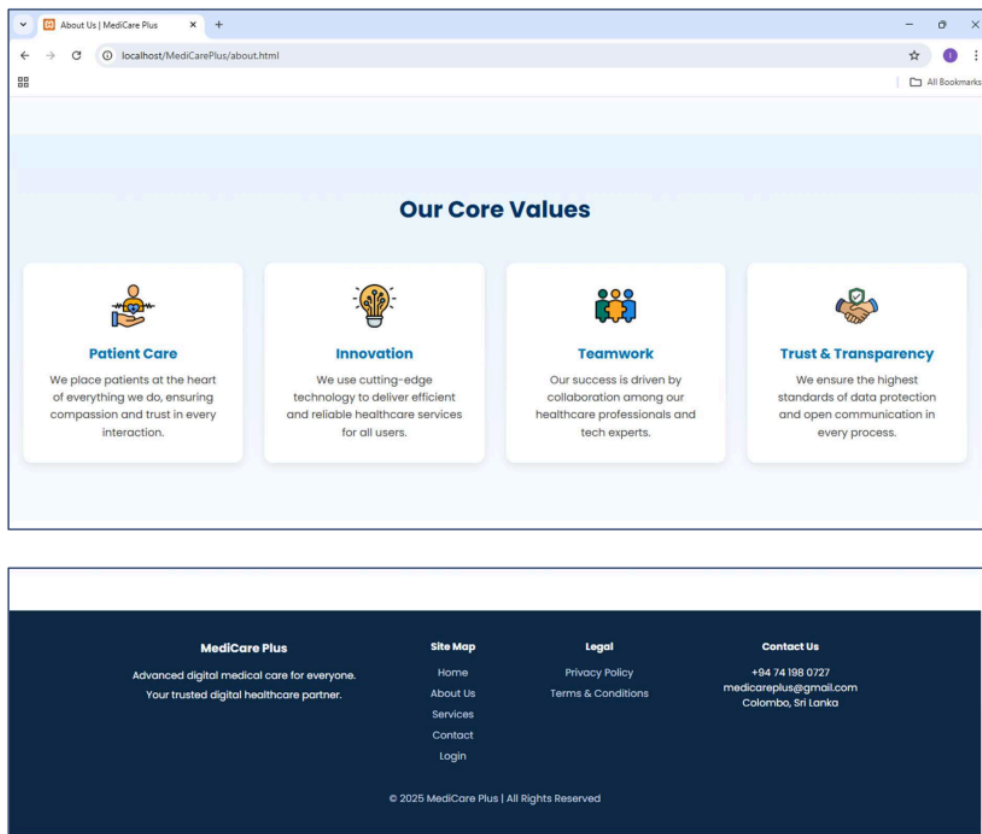




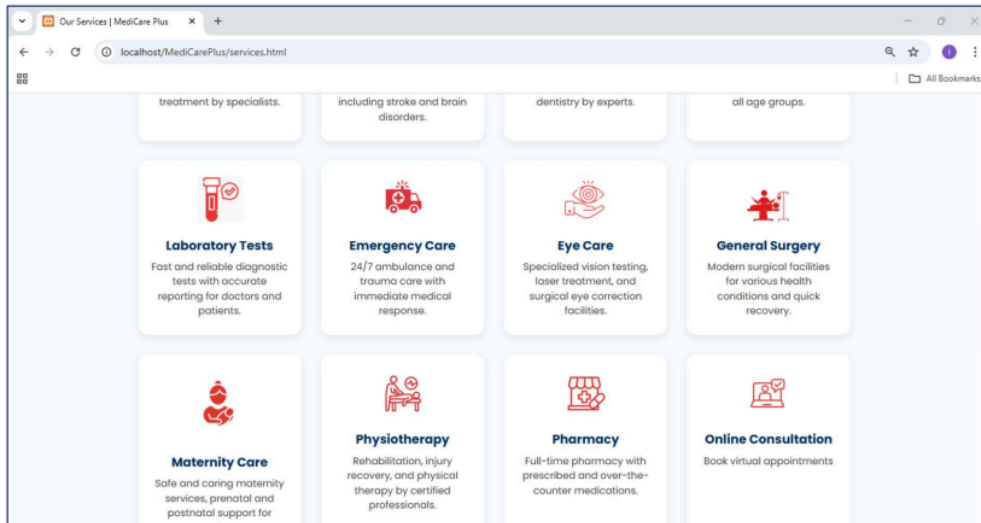
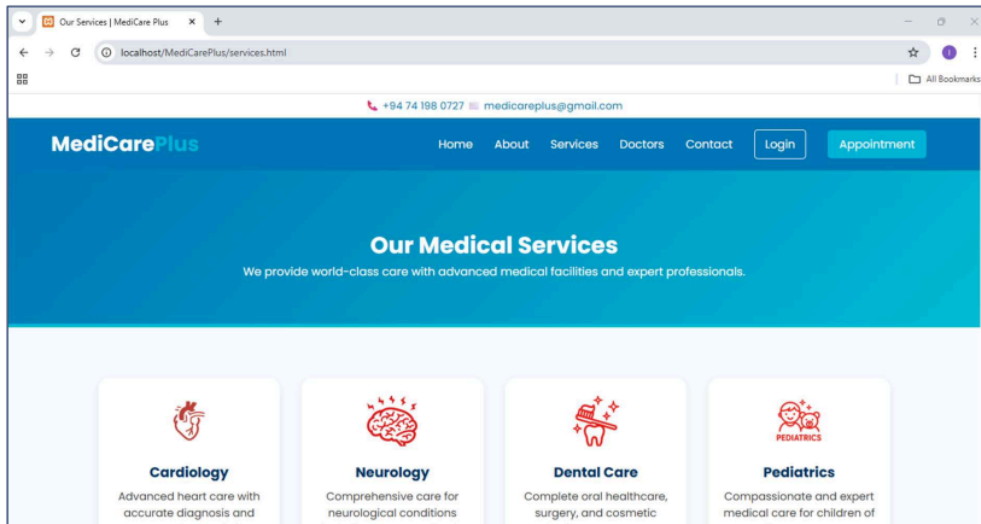


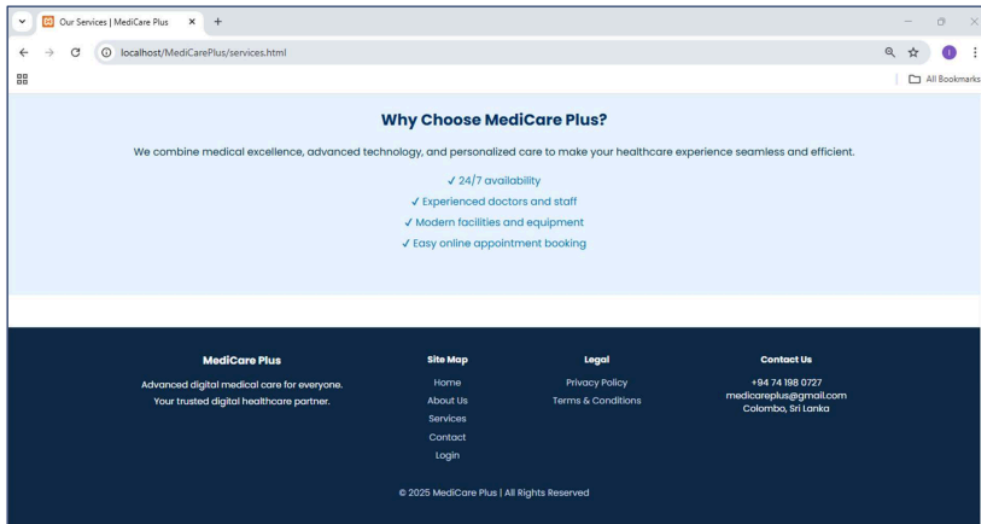
1. About



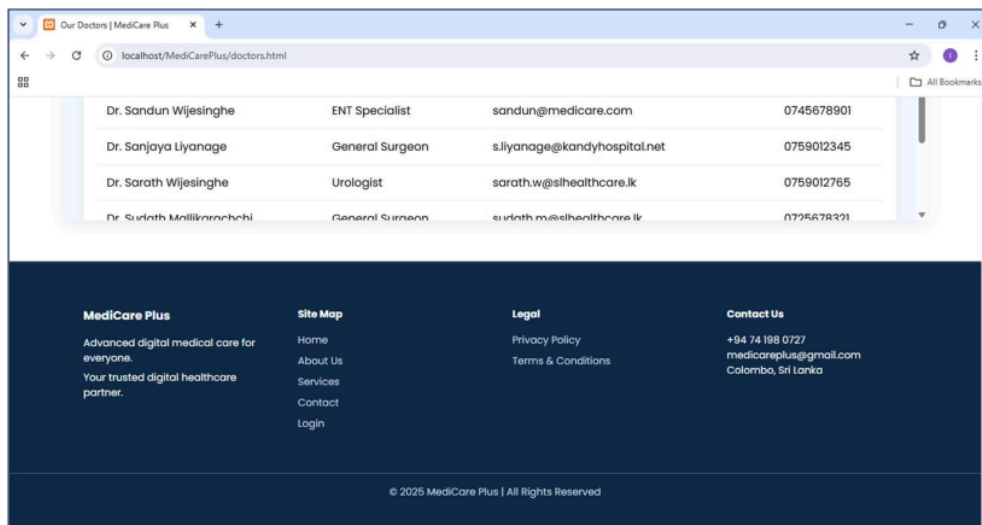
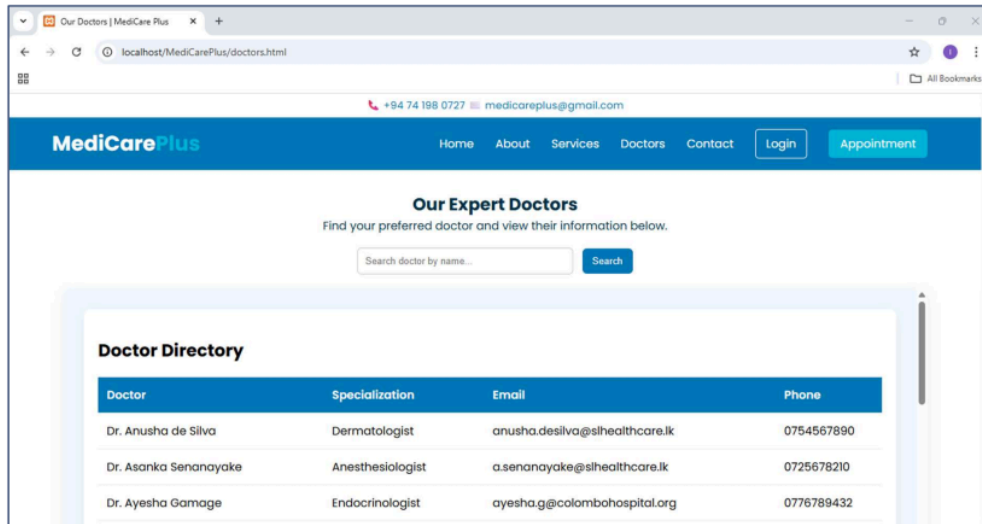


2. Services





3. Doctors



4. Contact

The screenshot shows a web browser window with the address bar displaying 'localhost/MediCarePlus/contact.html'. The page has a blue header with the 'MediCarePlus' logo and navigation links: Home, About, Services, Doctors, Contact, Login, and Appointment. Below the header, the 'Get in Touch' section contains contact details: Address: 120 Main Street, Colombo, Sri Lanka; Phone: +94 74 123 4567; Email: info@medicareplus.com; and Working Hours: Mon-Fri, 8:00 AM - 6:00 PM. To the right, the 'Send a Message' form includes input fields for 'Your Name', 'Your Email', 'Subject', and a text area for 'Your Message'.

MediCarePlus

Home About Services Doctors Contact Login Appointment

Get in Touch

Have questions or need support? Reach out to us using the form below.

Address: 120 Main Street, Colombo, Sri Lanka
Phone: +94 74 123 4567
Email: info@medicareplus.com
Working Hours: Mon-Fri, 8:00 AM - 6:00 PM

Send a Message

Your Name

Your Email

Subject

Your Message

This screenshot shows the same web browser window, but the 'Send Message' button is now visible below the message form. The footer is a dark blue section with four columns: 'MediCare Plus' (Advanced digital medical care for everyone. Your trusted digital healthcare partner.), 'Site Map' (Home, About Us, Services, Contact, Login), 'Legal' (Privacy Policy, Terms & Conditions), and 'Contact Us' (+94 74 188 0727, medicareplus@gmail.com, Colombo, Sri Lanka). The copyright notice '© 2025 MediCare Plus | All Rights Reserved' is at the bottom.

Send Message

MediCare Plus
Advanced digital medical care for everyone.
Your trusted digital healthcare partner.

Site Map
Home
About Us
Services
Contact
Login

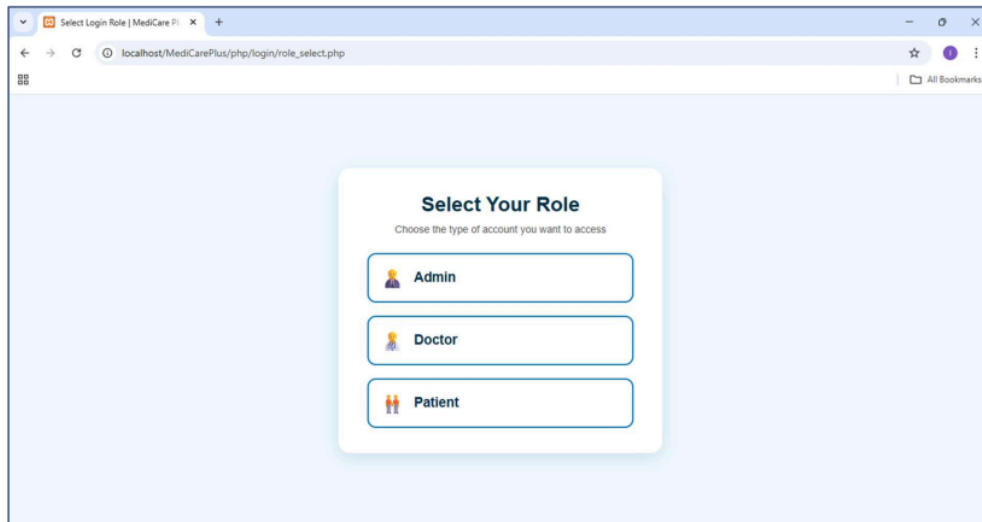
Legal
Privacy Policy
Terms & Conditions

Contact Us
+94 74 188 0727
medicareplus@gmail.com
Colombo, Sri Lanka

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5. Login

- Role Selection



- Login Page

Admin Login | MediCare Plus

localhost/MediCarePlus/php/login/login.php?role=admin

Admin Login

Enter your credentials to continue

Email

admin@gmail.com

Password

...

Login

Patient Login | MediCare Plus

localhost/MediCarePlus/php/login/login.php?role=patient

Patient Login

Enter your credentials to continue

Email

admin@gmail.com

Password

...

Login

Don't have an account? [Sign up](#)

- Signup Page

The screenshot shows a web browser window with the address bar displaying 'localhost/MediCarePlus/php/login/signup.php'. The page features a central white card with the title 'Create Patient Account' and a subtitle 'Fill in the details to register'. The card contains four input fields: 'Full Name', 'Email' (pre-filled with 'admin@gmail.com'), 'Phone', and 'Password' (with a masked input). Below these fields is a blue 'Create Account' button and a link that says 'Already have an account? [Login](#)'.

6. Appointment booking

The screenshot shows a web browser window with the address bar displaying 'localhost/MediCarePlus/php/appointment.php'. The page has a blue header with the 'MediCarePlus' logo and navigation links: 'Home', 'About', 'Services', 'Doctors', 'Contact', 'Login', and 'Appointment'. Below the header, a white card titled 'Book an Appointment' with the subtitle 'Please fill out the form below to schedule your visit.' contains a form with five fields: 'Your Full Name', 'Your Email', 'Your Phone Number', 'Select Doctor' (a dropdown menu), and a date field labeled 'mm/dd/yyyy' with a calendar icon.

Book Appointment | MediCare

localhost/MediCarePlus/php/appointment.php

Book an Appointment

Please fill out the form below to schedule your visit.

Your Full Name

Your Email

Your Phone Number

Select Doctor

mm/dd/yyyy

--:-- --

Additional Notes (Optional)

Book Appointment

Book Appointment | MediCare

localhost/MediCarePlus/php/appointment.php

Additional Notes (Optional)

Book Appointment

MediCare Plus

Advanced digital medical care for everyone.

Site Map

[Home](#)

[About Us](#)

[Contact](#)

[Login](#)

Legal

[Privacy Policy](#)

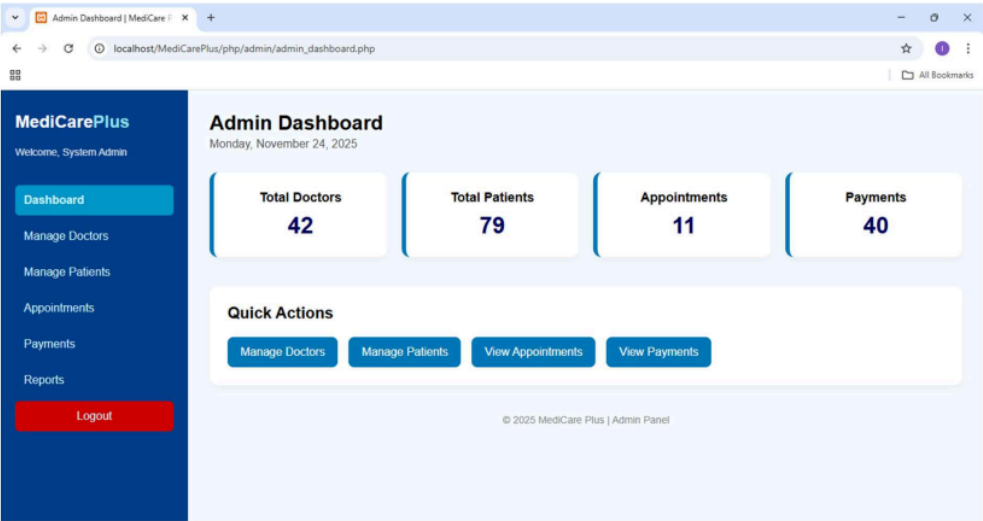
[Terms & Conditions](#)

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Admin Dashboard Pages

Admin pages maintain a formal and structural look, reflecting system-wide authority. The sidebar gives immediate access to all management areas: doctors, patients, payments, and appointments. Dashboards provide clear statistical overviews for decision-making. Tables and forms follow a consistent design to ensure accuracy and reduce administrative errors.

1. Manage doctors



2. Manage patients

Manage Patients | Admin Panel

localhost/MediCarePlus/php/admin/manage_patients.php

MediCarePlus

Welcome, System Admin

Dashboard
Manage Doctors
Manage Patients
Appointments
Payments
Reports
Logout

Manage Patients

ID	Patient Name	Email	Phone	Actions
1	Nimal Perera	nimalp01@gmail.com	0771234567	Delete
2	Kavindi Silva	kavindis@yahoo.com	0712345678	Delete
3	Tharindu Jayasinghe	tharindu.j@gmail.com	0756543210	Delete
4	Sachini Fernando	sachini.f@hotmail.com	0769988776	Delete
5	Malith Gunasekara	malithg@gmail.com	0778822114	Delete
6	Dinithi Weerasinghe	dinithi.w@gmail.com	0704455667	Delete
7	Isuru Ranasinghe	isurur@yahoo.com	0716655443	Delete
8	Senuri Madushani	senuri.m@gmail.com	0729856321	Delete
9	Pasindu Rathnayake	pasindur@gmail.com	0745621987	Delete
10	Bimsara Alwis	bimsaraa@gmail.com	0754412233	Delete

3. Manage doctors

Manage Doctors | MediCare Plus

localhost/MediCarePlus/php/admin/manage_doctors.php

MediCarePlus

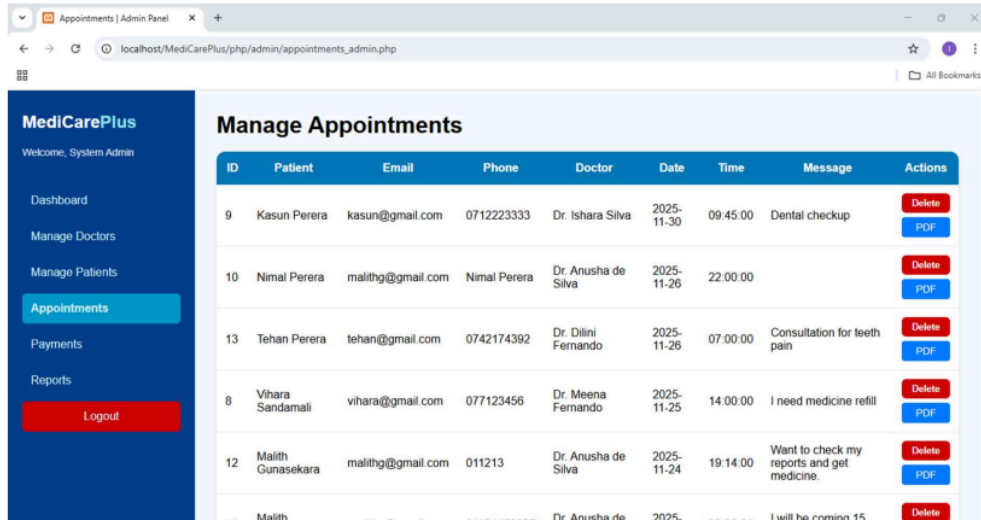
Welcome, System Admin

Dashboard
Manage Doctors
Manage Patients
Appointments
Payments
Reports
Logout

Manage Doctors

ID	Name	Specialization	Email	Phone	Actions
46	Indika Lakshmi	Dentist	indika@gmail.com	077917592	Delete
44	Dr. Malini Seneviratne	Radiologist	malini.s@kandyhospital.net	0720123876	Delete
43	Dr. Sarath Wijesinghe	Urologist	sarath.w@sihealthcare.lk	0759012765	Delete
42	Dr. Indika Alwis	ENT Specialist	i.alwis@medicare.com	0768901654	Delete
40	Dr. Ayesha Gamage	Endocrinologist	ayesha.g@colombohospital.org	0776789432	Delete
39	Dr. Sudath Mallikarachchi	General Surgeon	sudath.m@sihealthcare.lk	0725678321	Delete
38	Dr. Madhavi Tennakoon	Gynecologist	m.tennakoon@medicare.com	0754567210	Delete
37	Dr. Janith Weerakoon	Neurologist	janith.w@kandyhospital.net	0763456109	Delete
36	Dr. Upeksha Peiris	Psychiatrist	u.peiris@sihealthcare.lk	0712345098	Delete
35	Dr. Chaminda Dissanayake	Orthopedic Surgeon	c.dissanayake@colombohospital.org	0771234987	Delete

4. Appointments



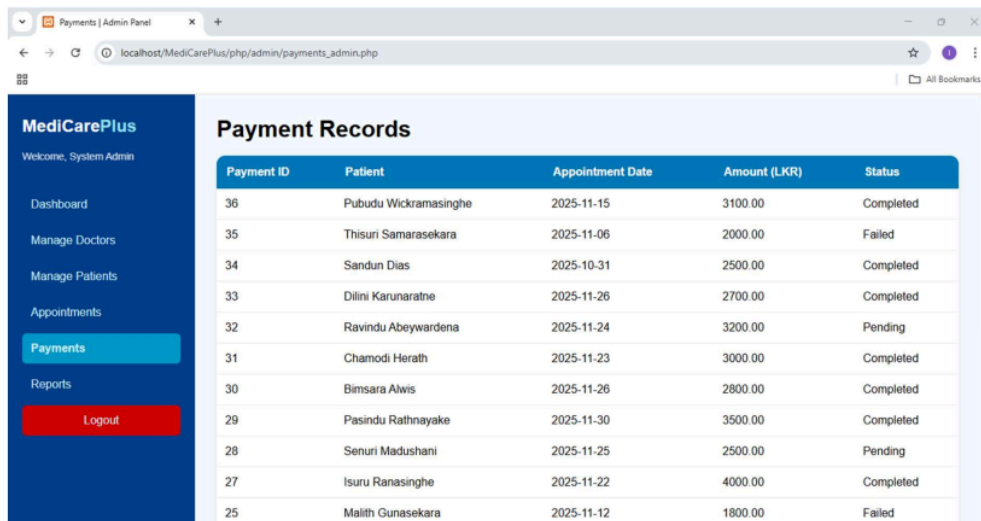
MediCarePlus
Welcome, System Admin

Dashboard
Manage Doctors
Manage Patients
Appointments
Payments
Reports
Logout

Manage Appointments

ID	Patient	Email	Phone	Doctor	Date	Time	Message	Actions
9	Kasun Perera	kasun@gmail.com	0712223333	Dr. Ishara Silva	2025-11-30	09:45:00	Dental checkup	Delete PDF
10	Nimal Perera	malith@gmail.com	Nimal Perera	Dr. Anusha de Silva	2025-11-26	22:00:00		Delete PDF
13	Tehan Perera	tehan@gmail.com	0742174392	Dr. Dilini Fernando	2025-11-26	07:00:00	Consultation for teeth pain	Delete PDF
8	Vihara Sandamali	vihara@gmail.com	077123456	Dr. Meena Fernando	2025-11-25	14:00:00	I need medicine refill	Delete PDF
12	Malith Gunasekara	malith@gmail.com	011213	Dr. Anusha de Silva	2025-11-24	19:14:00	Want to check my reports and get medicine.	Delete PDF
..	Malith			Dr. Anusha de	2025-		I will be coming 15	Delete

5. Payments



MediCarePlus
Welcome, System Admin

Dashboard
Manage Doctors
Manage Patients
Appointments
Payments
Reports
Logout

Payment Records

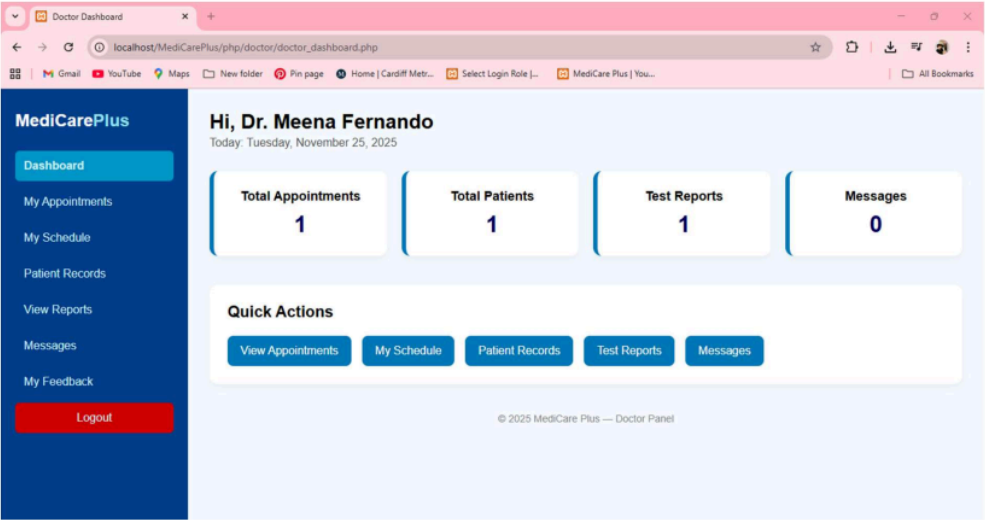
Payment ID	Patient	Appointment Date	Amount (LKR)	Status
36	Pubudu Wickramasinghe	2025-11-15	3100.00	Completed
35	Thisuri Samarasekara	2025-11-06	2000.00	Failed
34	Sandun Dias	2025-10-31	2500.00	Completed
33	Dilini Karunaratne	2025-11-26	2700.00	Completed
32	Ravindu Abeywardena	2025-11-24	3200.00	Pending
31	Chamodi Herath	2025-11-23	3000.00	Completed
30	Bimsara Alwis	2025-11-26	2800.00	Completed
29	Pasindu Rathnayake	2025-11-30	3500.00	Completed
28	Senuri Madushani	2025-11-25	2500.00	Pending
27	Isuru Ranasinghe	2025-11-22	4000.00	Completed
25	Malith Gunasekara	2025-11-12	1800.00	Failed

6. Reports

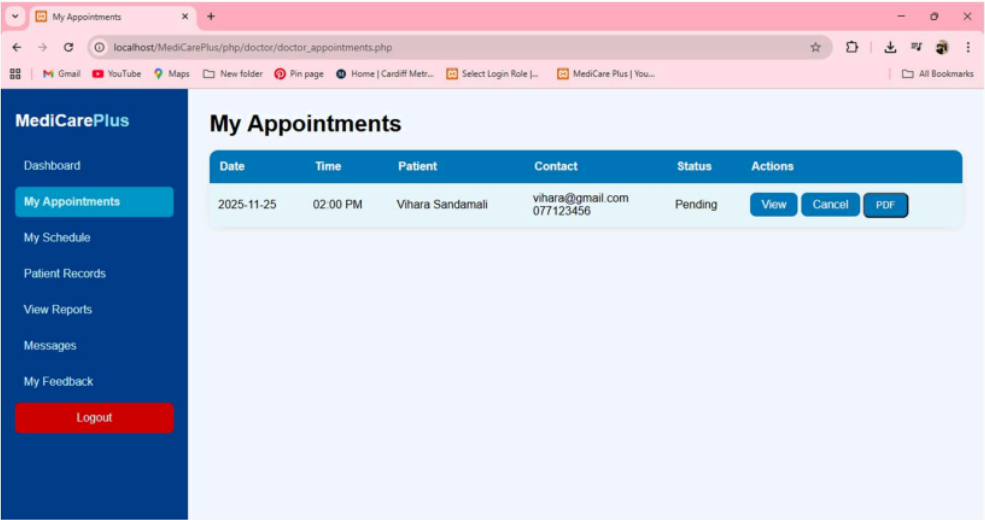
ID	Patient	Doctor	Type	Uploaded	Comments	Download
6	Malith Gunasekara	Dr. Meena Fernando	ss	2025-11-25 00:48:01	sss	PDF
1	Nimal Perera	Dr. Tharindu Jayasinghe	Chest X-Ray	2025-11-21 10:34:35	Suspected pneumonia — follow-up recommended	PDF
2	Kavindi Silva	Dr. Kavindi Ranasinghe	Blood Test - CBC	2025-11-21 10:34:35	Elevated WBC, possible infection	PDF
3	Tharindu Jayasinghe	Dr. Roshan Fernando	ECG	2025-11-21 10:34:35	Normal sinus rhythm, review in 6 months	PDF
5	Malith Gunasekara	Dr. Nuwan Perera		0000-00-00 00:00:00	2025/22/10	PDF

Doctor Dashboard Pages

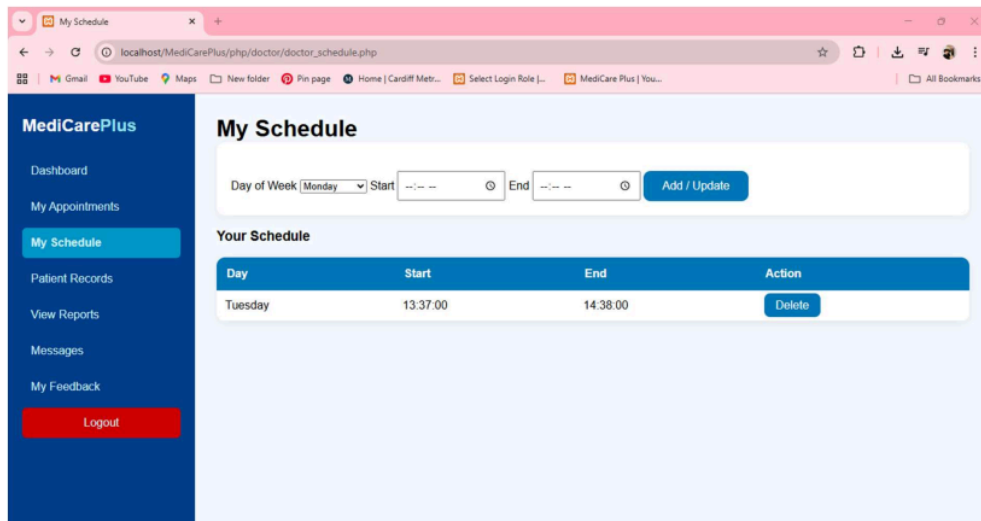
Doctor pages focus on efficiency and clarity. The dashboard highlights total appointments, patients, reports, and messages to help doctors plan their daily workflow. Sections like “Patient records”, “Test records”, and “Schedule” are organized for quick access. The system enables doctors to upload reports, issue prescriptions, and respond to patient messages with minimal clicks.



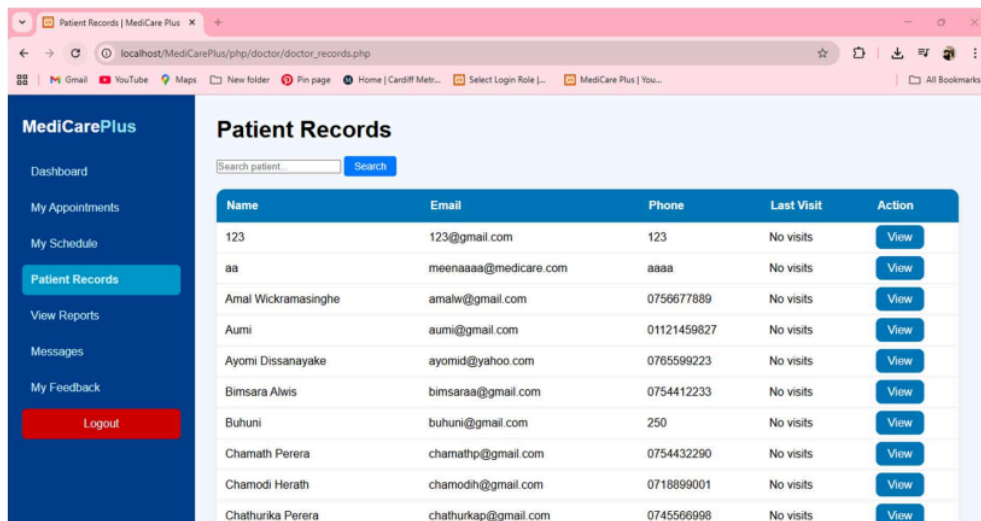
1. My Appointments



2. My Schedule



3. Patient records



4. View Reports

Medical Reports | MediCare Plus

localhost/MediCarePlus/php/doctor/doctor_reports.php

MediCarePlus

Dashboard

My Appointments

My Schedule

Patient Records

View Reports

Messages

My Feedback

Logout

Test Reports

Upload New Test Report

Patient Email

Enter patient email

Report Type

Blood Test, ECG, Scan, X-Ray...

Upload PDF Report

Choose File No file chosen

Comments

Optional notes...

Upload Report

5. Messages

Messages | Doctor

localhost/MediCarePlus/php/doctor/doctor_messages.php

MediCarePlus

Dashboard

My Appointments

My Schedule

Patient Records

Reports

Messages

Logout

Messages

Send Message to a Patient

Patient Email

Message

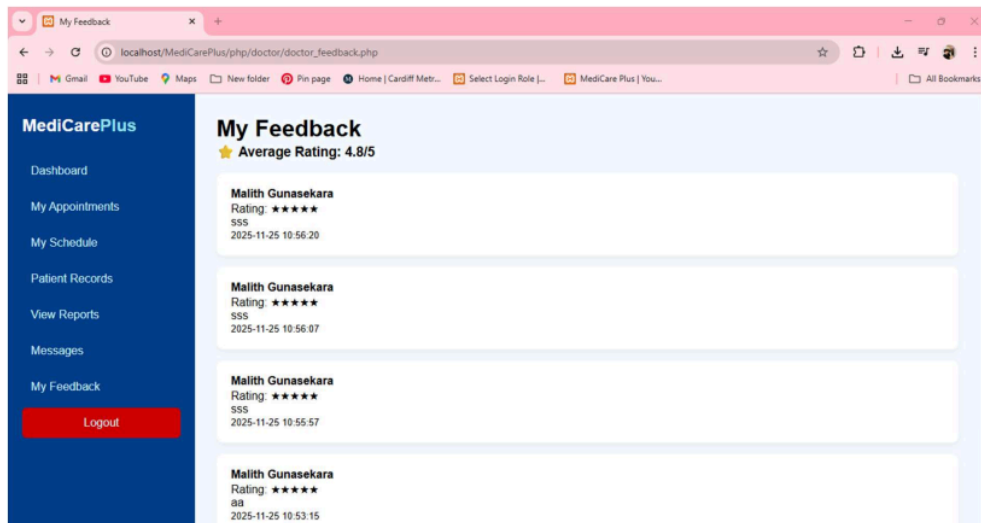
Send

Inbox (Patients -> You)

Patient: malithg@gmail.com
2025-11-24 19:47:18
Did you check my reports??

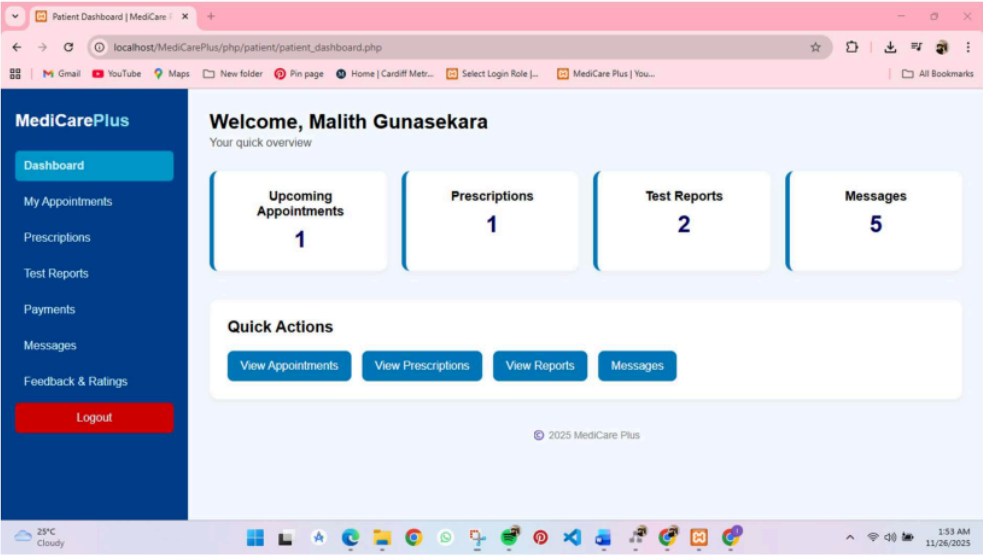
Patient: malithg@gmail.com
2025-11-24 19:47:02
Did

6. My Feedback

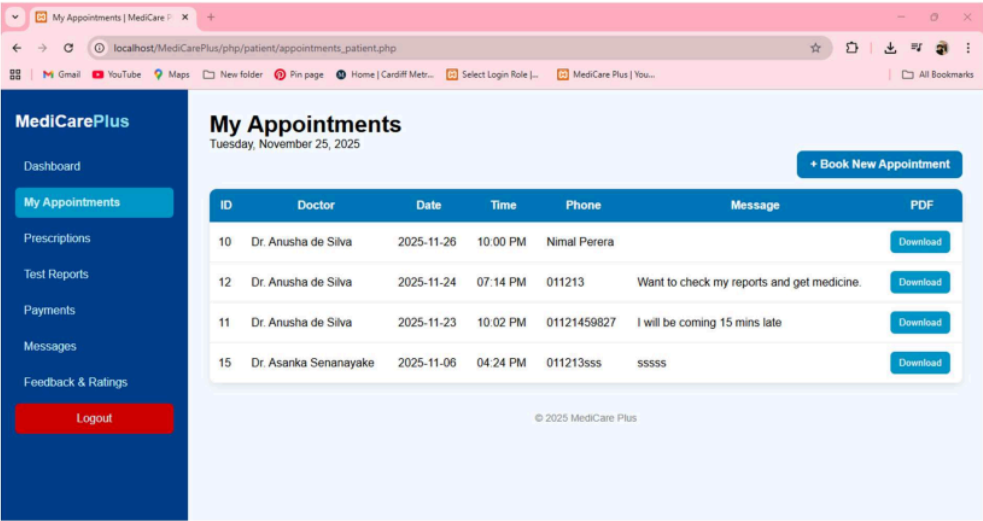


Patient Dashboard Pages

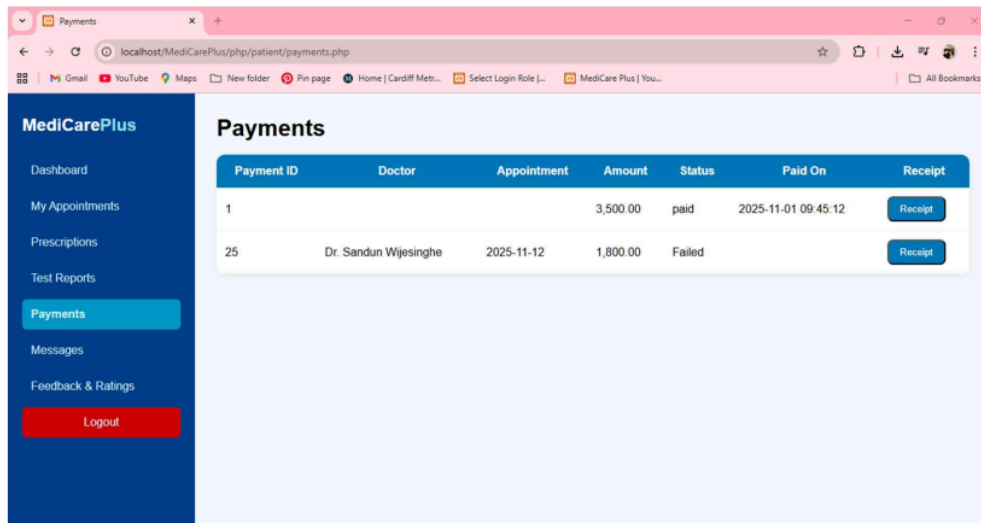
Every patient page is designed to be simple, clean, and easy to navigate. The dashboard provides a quick overview of upcoming appointments, prescriptions, reports, payments, and messages. Large buttons and card-based layouts help users quickly locate required actions such as booking appointments or checking results. Colors are soft and calm to align with healthcare themes, ensuring a stress-free experience.



1. My Appointments



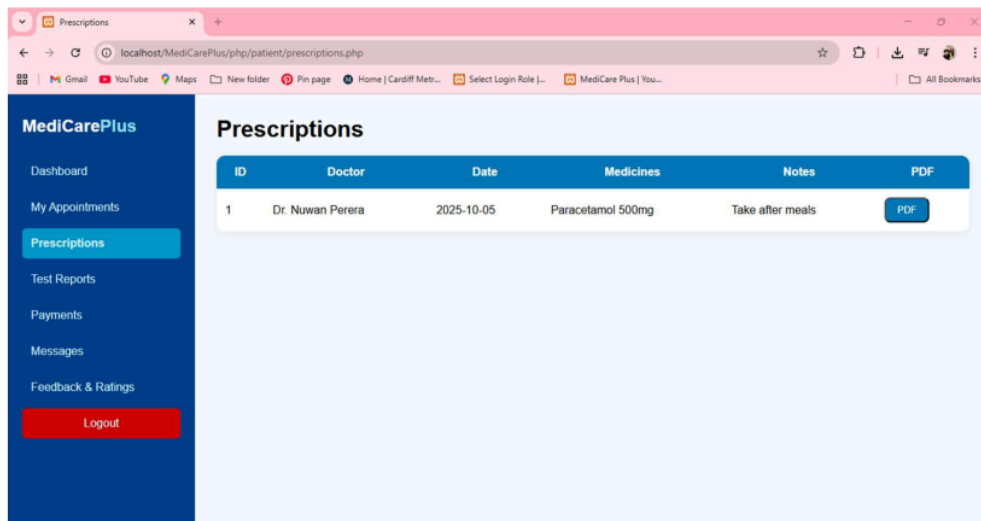
2. Payments



The screenshot shows the 'Payments' page of the MediCarePlus application. The left sidebar contains navigation links: Dashboard, My Appointments, Prescriptions, Test Reports, Payments (highlighted), Messages, Feedback & Ratings, and a Logout button. The main content area is titled 'Payments' and displays a table with the following data:

Payment ID	Doctor	Appointment	Amount	Status	Paid On	Receipt
1			3,500.00	paid	2025-11-01 09:45:12	Receipt
25	Dr. Sandun Wijesinghe	2025-11-12	1,800.00	Failed		Receipt

3. Prescriptions



The screenshot shows the 'Prescriptions' page of the MediCarePlus application. The left sidebar contains navigation links: Dashboard, My Appointments, Prescriptions (highlighted), Test Reports, Payments, Messages, Feedback & Ratings, and a Logout button. The main content area is titled 'Prescriptions' and displays a table with the following data:

ID	Doctor	Date	Medicines	Notes	PDF
1	Dr. Nuwan Perera	2025-10-05	Paracetamol 500mg	Take after meals	PDF

4. Test Reports

The screenshot shows the 'My Test Reports' page in the MediCarePlus application. The browser address bar indicates the URL is `localhost/MediCarePlus/php/patient/reports.php`. The left sidebar contains a menu with the following items: Dashboard, My Appointments, Prescriptions, Test Reports (highlighted), Payments, Messages, Feedback & Ratings, and a Logout button. The main content area is titled 'My Test Reports' with the date 'Tuesday, November 25, 2025'. It displays a table with the following data:

ID	Type	Doctor	Uploaded	Comments	PDF
6	ss	Dr. Meena Fernando	2025-11-25 00:48:01	sss	Download
5		Dr. Nuwan Perera	0000-00-00 00:00:00	2025/22/10	Download

Below the table, the copyright notice '© 2025 MediCare Plus' is visible.

5. Feedback and Ratings

The screenshot shows the 'Feedback & Ratings' page in the MediCarePlus application. The browser address bar indicates the URL is `localhost/MediCarePlus/php/patient/patient_feedback.php`. The left sidebar is identical to the previous page, with 'Feedback & Ratings' highlighted. The main content area is titled 'Feedback & Ratings' and contains a form with the following fields:

- Select Doctor:** A dropdown menu with 'Choose' selected.
- Rating:** A row of five stars, with the first three stars filled.
- Comment:** A text input field with the placeholder text 'Your experience...'
- Submit Feedback:** A button at the bottom of the form.

6. Messages

The screenshot displays the 'Messages' page of the MediCarePlus application. On the left is a dark blue sidebar with navigation links: Dashboard, My Appointments, Prescriptions, Test Reports, Payments, Messages (highlighted in light blue), Feedback & Ratings, and a red Logout button. The main content area has a light blue header with the title 'Messages'. Below the header is a form to send a message, featuring a 'Send to:' dropdown menu with 'Doctor' selected, a 'Receiver email' input field, a 'Message' input field, and a blue 'Send' button. Underneath the form is a section titled 'Recent Messages' containing a list of five messages. Each message entry shows the sender's email, the receiver's email, the message content, and the timestamp.

From	To	Message	Time
patient.malith@gmail.com	meena@medicare.com	Did you check my reports??	2025-11-24 19:47:18
patient.malith@gmail.com	meena@medicare.com	Did	2025-11-24 19:47:02
patient.malith@gmail.com	meena@medicare.com	S	2025-11-24 16:52:55
patient.malith@gmail.com	nuvan@medicare.com	SSS	2025-11-23 10:41:17
patient.malith@gmail.com	nuvan@medicare.com	SSS	2025-11-23 10:41:10

a) Implementation of backend

The backend is implemented using PHP and MySQL.

1. System Architecture

Three-tier architecture.

Layer	Description
13 Presentation Layer	HTML/CSS/UI
Application Layer	PHP for business logic
Data Layer	MySQL database storing all records

2. Key Backend Features Implemented

1) Secure Login System (Admin/Doctor/Patient)

- Passwords stored using a password_hash()
- Secure validation using password_verify()
- Role-based session variables:

Example:

- Redirects based on role:

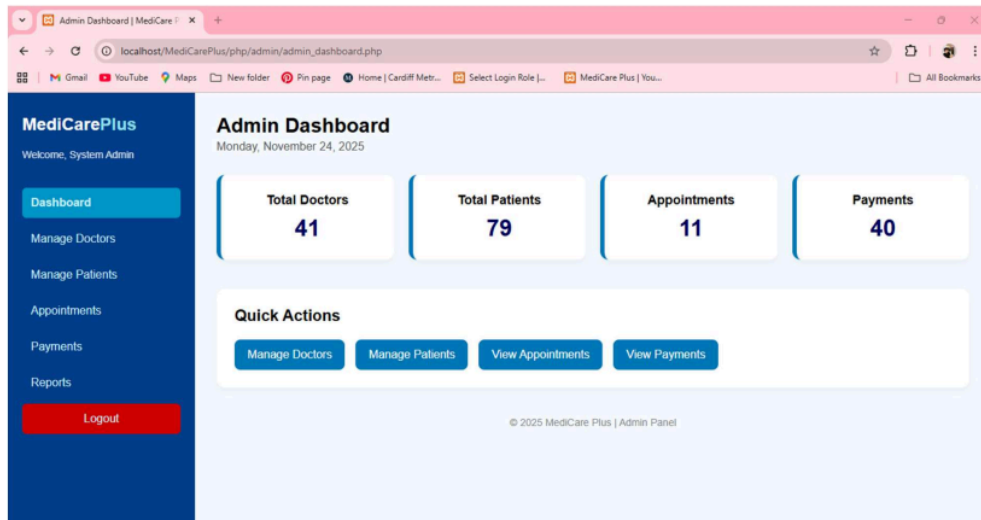
Example:

2) Secure Signup System

- Input validation
- Email uniqueness check
- Automatic session starts after signup
- User added to proper table.

3) Dashboard Functionalities

Admin Side



Features	Implementation Summary
Manage doctors	CRUD operations on the doctors' table.
Manage patients	CRUD operations on the patient's table.
Appointments	View/edit/delete
Payments	View payment records
Reports	PHP – MySQL queries, printable pages
Logout	Session_destroy()

Doctor Side

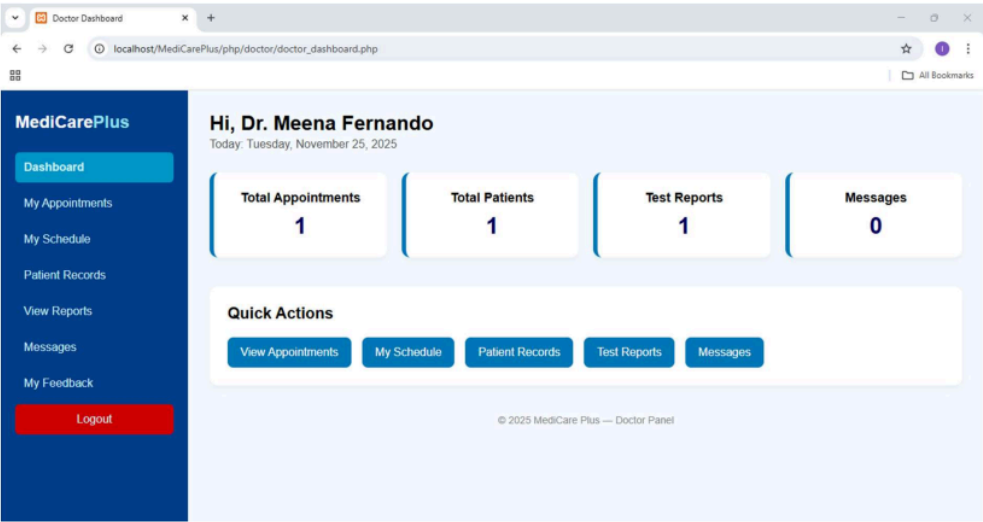
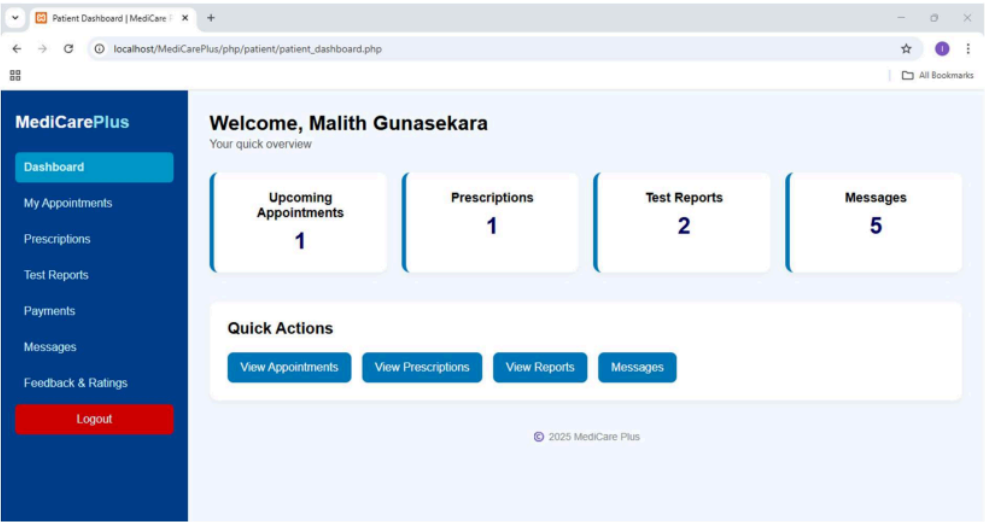


Figure 2Features	Implementation Summary
View appointments	SELECT queries filtered by doctor_id
Cancel appointments	UPDATE appointment SET status = “cancelled”
View patient records	JOIN appointments + patients
View uploaded reports	Fetch from the reports table
Messaging	Doctor ↔ patient communication
Schedule management	Insert/update availability

Patient Side

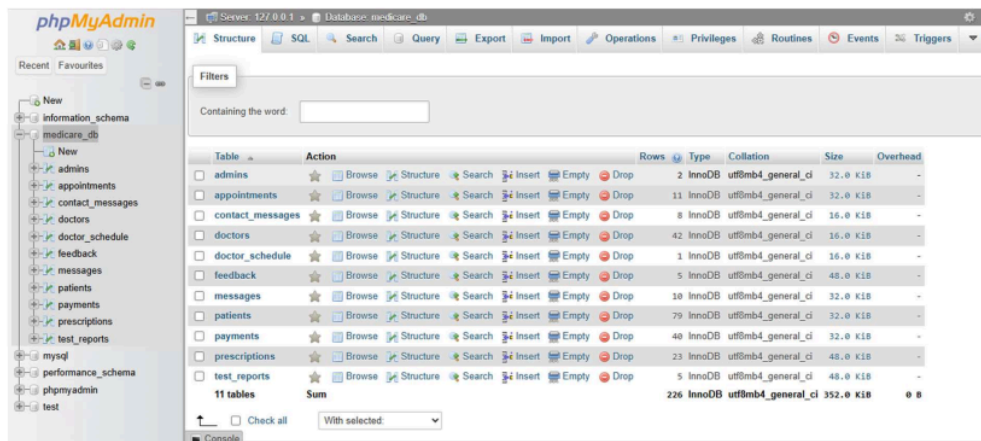


Features	Implementation Summary
Book appointment	Insert appointment record
View past appointments	WHERE patient_id = session_id
View prescriptions	Fetch from prescriptions table
Download reports	PDF
Give feedback	Insert into feedback table
Messaging	Secure private messaging

4) Security Measures Implemented

- SQL injection prevention - All database operations use prepared statements.
- Session Security
 - Role-based access
 - Session timeout can be enabled
 - No dashboard access without login
- Input Sanitization
 - trim()
 - filter_input()
- Password Encryption

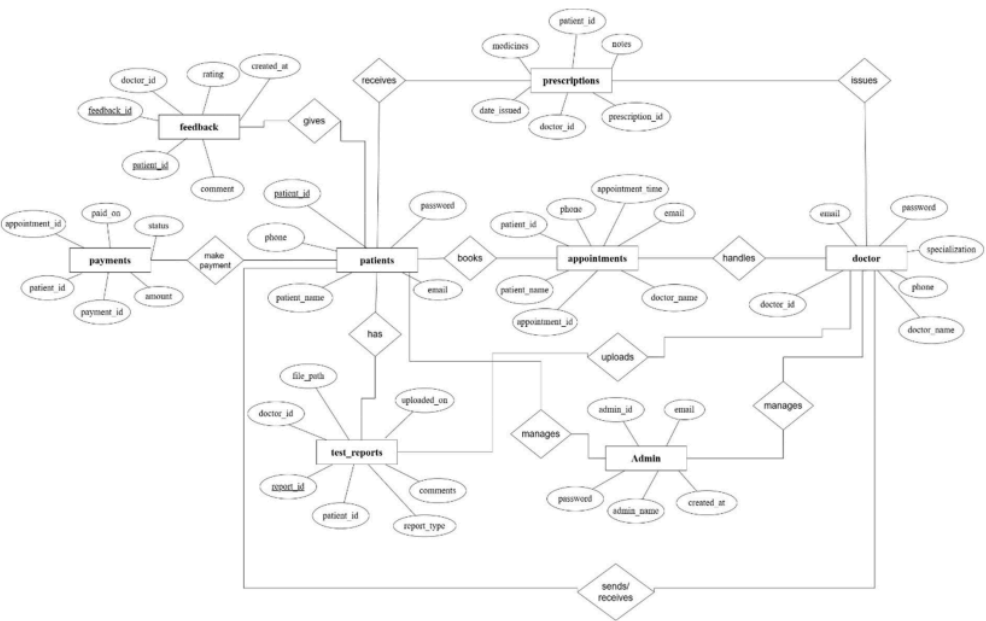
5) Database Design Summary



The screenshot shows the phpMyAdmin interface for a database named 'medicare_db'. The left sidebar displays a tree view of the database structure, including tables like 'admins', 'appointments', 'contact_messages', 'doctors', 'doctor_schedule', 'feedback', 'messages', 'patients', 'payments', 'prescriptions', 'test_reports', 'test', 'performance_schema', 'phpmyadmin', and 'mysql'. The main panel shows the 'Structure' tab for the 'medicare_db' database. It lists 11 tables with their respective actions (Browse, Structure, Search, Insert, Empty, Drop) and details such as Rows, Type, Collation, Size, and Overhead.

Table	Action	Rows	Type	Collation	Size	Overhead
admins	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	32.0 KiB	-
appointments	Browse Structure Search Insert Empty Drop	11	InnoDB	utf8mb4_general_ci	32.0 KiB	-
contact_messages	Browse Structure Search Insert Empty Drop	8	InnoDB	utf8mb4_general_ci	16.0 KiB	-
doctors	Browse Structure Search Insert Empty Drop	42	InnoDB	utf8mb4_general_ci	16.0 KiB	-
doctor_schedule	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 KiB	-
feedback	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	48.0 KiB	-
messages	Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	32.0 KiB	-
patients	Browse Structure Search Insert Empty Drop	79	InnoDB	utf8mb4_general_ci	32.0 KiB	-
payments	Browse Structure Search Insert Empty Drop	40	InnoDB	utf8mb4_general_ci	32.0 KiB	-
prescriptions	Browse Structure Search Insert Empty Drop	23	InnoDB	utf8mb4_general_ci	48.0 KiB	-
test_reports	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	48.0 KiB	-
11 tables	Sum	226	InnoDB	utf8mb4_general_ci	352.0 KiB	0 B

ER Diagram



Main tables:

- doctors

	doctor_id	doctor_name	specialization	email	phone	password
☐ Edit ☐ Copy ☐ Delete	1	Dr. Nuwan Perera	Cardiologist	nuwan@medicare.com	0712345678	nuwan
☐ Edit ☐ Copy ☐ Delete	2	Dr. Meena Fernando	Neurologist	meena@medicare.com	0723456789	meena
☐ Edit ☐ Copy ☐ Delete	3	Dr. Ishara Silva	Dentist	ishara@medicare.com	0756789012	NULL
☐ Edit ☐ Copy ☐ Delete	4	Dr. Tharindu Jayasinghe	Pediatrician	tharindu@medicare.com	0764567890	NULL
☐ Edit ☐ Copy ☐ Delete	5	Dr. Kavindi Ranasinghe	Dermatologist	kavindi@medicare.com	0773456789	NULL
☐ Edit ☐ Copy ☐ Delete	6	Dr. Roshan Fernando	Orthopedic Surgeon	roshan@medicare.com	0781234567	123
☐ Edit ☐ Copy ☐ Delete	7	Dr. Malithi Perera	Gynecologist	malithi@medicare.com	0709876543	NULL
☐ Edit ☐ Copy ☐ Delete	8	Dr. Sandun Wijesinghe	ENT Specialist	sandun@medicare.com	0745678901	NULL
☐ Edit ☐ Copy ☐ Delete	9	Dr. Hashini Silva	Psychiatrist	hashini@medicare.com	0717890123	NULL

- doctors_schedule

	schedule_id	doctor_id	day_of_week	start_time	end_time
☐ Edit ☐ Copy ☐ Delete	2	2	Tuesday	13:37:00	14:38:00
☐ Check all	With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export				

- messages

	message_id	sender_role	sender_email	sender_id	receiver_role	receiver_email	receiver_id	content	sent_on
☐ Edit ☐ Copy ☐ Delete	1	patient	0	NULL	doctor	0	NULL	did you check my reports?	2025-11-22 18:48:33
☐ Edit ☐ Copy ☐ Delete	2	patient	0	NULL	doctor	0	NULL	xx	2025-11-23 00:05:31
☐ Edit ☐ Copy ☐ Delete	3	patient	0	NULL	doctor	0	NULL	222	2025-11-23 00:23:48
☐ Edit ☐ Copy ☐ Delete	4	patient	0	NULL	doctor	0	NULL	MY REPORTS	2025-11-23 00:46:31
☐ Edit ☐ Copy ☐ Delete	5	patient	malithg@gmail.com	NULL	admin	nuwan@medicare.com	NULL	sss	2025-11-23 10:41:10
☐ Edit ☐ Copy ☐ Delete	6	patient	malithg@gmail.com	NULL	doctor	nuwan@medicare.com	NULL	sss	2025-11-23 10:41:17
☐ Edit ☐ Copy ☐ Delete	7	doctor	meena@medicare.com	NULL	patient	nuwan@medicare.com	NULL	I checked your reports	2025-11-24 04:51:36
☐ Edit ☐ Copy ☐ Delete	8	patient	malithg@gmail.com	NULL	doctor	meena@medicare.com	NULL	s	2025-11-24 10:52:55
☐ Edit ☐ Copy ☐ Delete	9	patient	malithg@gmail.com	NULL	doctor	meena@medicare.com	NULL	Did	2025-11-24 19:47:02
☐ Edit ☐ Copy ☐ Delete	10	patient	malithg@gmail.com	NULL	doctor	meena@medicare.com	NULL	Did you check my reports??	2025-11-24 19:47:18

- payments

	payment_id	patient_id	appointment_id	amount	status	paid_on
☐ Edit ☐ Copy ☐ Delete	1	5	4	3500.00	paid	2025-11-01 09:45:12
☐ Edit ☐ Copy ☐ Delete	2	12	5	4200.00	paid	2025-11-02 10:12:54
☐ Edit ☐ Copy ☐ Delete	3	8	6	2800.00	pending	NULL
☐ Edit ☐ Copy ☐ Delete	4	19	7	5000.00	paid	2025-11-03 14:22:19
☐ Edit ☐ Copy ☐ Delete	5	25	8	3200.00	paid	2025-11-04 16:10:44
☐ Edit ☐ Copy ☐ Delete	6	37	9	4500.00	pending	NULL
☐ Edit ☐ Copy ☐ Delete	7	14	10	3800.00	paid	2025-11-05 18:40:02
☐ Edit ☐ Copy ☐ Delete	8	41	11	6000.00	paid	2025-11-06 09:55:31
☐ Edit ☐ Copy ☐ Delete	9	2	12	2500.00	paid	2025-11-07 11:28:10

- prescriptions

	prescription_id	patient_id	doctor_id	date_issued	medicines	notes
☐ Edit ☐ Copy ☐ Delete	1	5	1	2025-10-05	Paracetamol 500mg	Take after meals
☐ Edit ☐ Copy ☐ Delete	2	12	3	2025-10-08	Amoxicillin 250mg	Finish full course
☐ Edit ☐ Copy ☐ Delete	3	22	8	2025-10-12	Ibuprofen 200mg	Take if pain persists
☐ Edit ☐ Copy ☐ Delete	4	9	15	2025-10-14	Vitamin C 1000mg	Take daily
☐ Edit ☐ Copy ☐ Delete	5	16	10	2025-10-16	Metformin 500mg	Monitor blood sugar
☐ Edit ☐ Copy ☐ Delete	6	7	18	2025-10-18	Cough Syrup 10ml	Use twice a day
☐ Edit ☐ Copy ☐ Delete	7	14	21	2025-10-20	Cetirizine 10mg	For allergies
☐ Edit ☐ Copy ☐ Delete	8	3	12	2025-10-22	Aspirin 81mg	Take with water
☐ Edit ☐ Copy ☐ Delete	9	19	4	2025-10-25	Lisinopril 10mg	Blood pressure
☐ Edit ☐ Copy ☐ Delete	10	27	13	2025-10-26	Azithromycin 500mg	1 tablet for 3 days
☐ Edit ☐ Copy ☐ Delete	11	41	20	2025-10-29	Insulin 10 units	Before meals
☐ Edit ☐ Copy ☐ Delete	12	32	12	2025-11-01	Losartan 50mg	Daily
☐ Edit ☐ Copy ☐ Delete	13	6	6	2025-11-03	Paracetamol 500mg	Take daily

• test_reports

#		report_id	patient_id	doctor_id	report_type	file_path	uploaded_on	comments	
☐				1	1	4 Chest X-Ray	/mnt/data/lobhdsce500pstsarwrtf1nov-2024mar202...	2025-11-21 10:34:35	Suspected pneumonia — follow-up recommended
☐				2	2	5 Blood Test - CBC	uploads/reports/cbc_report_2.pdf	2025-11-21 10:34:35	Elevated WBC, possible infection
☐				3	3	6 ECG	uploads/reports/ecg_report_3.pdf	2025-11-21 10:34:35	Normal sinus rhythm, review in 6 months
☐				5	5	1 ECG		0000-00-00 00:00:00	2025/22/10
☐				6	5	2 ss	uploads/reports/1764011881_report_1 (2).pdf	2025-11-25 00:48:01	sss

• patients

				patient_id	patient_name	email	phone	password	created_at
☐				1	Nimal Perera	nimalp01@gmail.com	0771234567	123	2025-11-23 11:24:21
☐				2	Kavindi Silva	kavindis@yahoo.com	0712345678	kavindi	2025-11-23 11:24:21
☐				3	Tharindu Jayasinghe	tharindu.j@gmail.com	0756543210	tharu123	2025-11-23 11:24:21
☐				4	Sachini Fernando	sachini.f@hotmail.com	0769988776	sachini	2025-11-23 11:24:21
☐				5	Malith Gunasekara	malithg@gmail.com	0778822114	malith	2025-11-23 11:24:21
☐				6	Dinithi Weerasinghe	dinithi.w@gmail.com	0704455667	dinithi	2025-11-23 11:24:21
☐				7	Isuru Ranasinghe	isurur@yahoo.com	0716655443	isuru	2025-11-23 11:24:21
☐				8	Senuri Madushani	senuri.m@gmail.com	0729856321	senuri	2025-11-23 11:24:21
☐				9	Pasindu Rathnayake	pasindur@gmail.com	0745621987	pasindu	2025-11-23 11:24:21
☐				10	Bimsara Alwis	bimsaraa@gmail.com	0754412233	bimsara	2025-11-23 11:24:21
☐				11	Chamodi Herath	chamodih@gmail.com	0718899001	chamodi	2025-11-23 11:24:21
☐				12	Ravindu Abeywardena	ravindu.a@yahoo.com	0763322115	ravindu	2025-11-23 11:24:21
☐				13	Dilini Karunaratne	dilini.k@gmail.com	0779900123	dilini	2025-11-23 11:24:21
☐				14	Sandun Dias	sandund@gmail.com	0745522109	sandun	2025-11-23 11:24:21

• admin

←T→		admin_id	admin_name	email	password	created_at
☐	Edit Copy Delete	1	System Admin	admin@gmail.com	123	2025-11-18 23:48:52
☐	Edit Copy Delete	2	System Admin	admin1@gmail.com	123	2025-11-18 23:49:44

• appointments

				appointment_id	patient_name	email	patient_id	phone	doctor_name	appointment_date	appointment_time	message	submitted_at	status
☐				Click the drop-down arrow to toggle column's visibility.	Sarah Perera	Sarah@gmail.com	1	011217895	Dr. Sandun Wijesinghe	2025-11-12	02:00:00	I need to know about the report results.	2025-11-09 00:48:13	pending
☐					Sarah Perera	sarah@gmail.com	2	074155625	Dr. Nuwan Perera	2025-11-22	10:30:00	Routine checkup	2025-11-20 08:41:00	pending
☐					Vihara Sandamal	vihara@gmail.com	3	077123456	Dr. Meena Fernando	2025-11-28	14:00:00	I need medicine refill	2025-11-21 16:30:00	pending
☐					Kasun Perera	kasun@gmail.com	NULL	0712223333	Dr. Ishara Silva	2025-11-30	09:45:00	Dental checkup	2025-11-21 18:22:00	pending
☐					Nimal Perera	malithg@gmail.com	5	Nimal Perera	Dr. Anusha de Silva	2025-11-26	22:00:00		2025-11-22 18:59:08	pending
☐				Malith Gunasekara	malithg@gmail.com	5	01121456827	Dr. Anusha de Silva	2025-11-23	22:02:00	I will be coming 15 mins late	2025-11-22 19:01:50	pending	
☐				Malith Gunasekara	malithg@gmail.com	5	011213	Dr. Anusha de Silva	2025-11-24	19:14:00	Want to check my reports and get medicine.	2025-11-22 23:10:02	pending	
☐				Tahan Perera	tahan@gmail.com	NULL	0742174362	Dr. Dinithi Fernando	2025-11-26	07:00:00	Consultation for tooth pain	2025-11-23 01:31:00	pending	
☐				Fah	tesh@gmail.com	NULL	01121456827	Dr. Asanka Senarathne	2025-10-31	11:47:00	5555	2025-11-23 11:42:42	pending	
☐				Malith Gunasekara	malithg@gmail.com	NULL	011213sss	Dr. Asanka Senarathne	2025-11-08	16:24:00	ssss	2025-11-23 22:19:58	pending	
☐				s	s@gmail.com	NULL	s	Dr. Kumari Silva	2025-11-16	19:58:00	22	2025-11-24 18:56:54	pending	

• contact_messages

				id	name	email	subject	message	submitted_at
☐				5	Hiruni Sawbhagya	hiruni@gmail.com	ECG	How much for ECG rept?	2025-10-30 12:02:14
☐				6	Vihara	vihara@gmail.com	Vaccines available?	Is vaccinations for children available.	2025-10-30 22:50:15
☐				11	David Matha	david@gmail.com	Vaccines available?	Price list please	2025-11-02 10:24:56
☐				12	Sithum Bandara	sithumb@gmail.com	I need report about my scan	Last week i took scan for that im requesting a rep...	2025-11-02 13:05:41
☐				13	Ann	david@gmail.com	sss	sssss	2025-11-14 22:38:42
☐				14	Ann	david@gmail.com	sss	sssss	2025-11-14 22:38:45
☐				15	Ann	david@gmail.com	sss	sssss	2025-11-14 22:42:16
☐				16	David Matha	david@gmail.com	I need report about my scan	ss	2025-11-23 01:58:27

- feedback

				feedback_id	patient_id	doctor_id	rating	comment	created_at
☐					1	5	2	4 ddd	2025-11-25 10:49:46
☐					2	5	2	5 aa	2025-11-25 10:53:15
☐					3	5	2	5 sss	2025-11-25 10:55:57
☐					4	5	2	5 sss	2025-11-25 10:56:07
☐					5	5	2	5 sss	2025-11-25 10:56:20

Relationships:

- 1¹⁸ Doctor – Many Appointments
- 1 Patient – Many Appointments
- 1 Appointment – Many Medical Reports

Table Name	Query
Admin	<pre>CREATE TABLE admins (admin_id INT(11) NOT NULL AUTO_INCREMENT, admin_name VARCHAR(100) NOT NULL, email VARCHAR(120) NOT NULL, password VARCHAR(255) NOT NULL, created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP, PRIMARY KEY (admin_id));</pre>
Patient	<pre>CREATE TABLE patients (patient_id INT(11) NOT NULL AUTO_INCREMENT, patient_name VARCHAR(100) NOT NULL, email VARCHAR(150) NOT NULL, phone VARCHAR(30) NOT NULL, password VARCHAR(255) NOT NULL, created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP, PRIMARY KEY (patient_id));</pre>
Doctors	<pre>CREATE TABLE doctors (doctor_id INT(11) NOT NULL AUTO_INCREMENT, doctor_name VARCHAR(100) NOT NULL, specialization VARCHAR(100), email VARCHAR(100), phone VARCHAR(20), password VARCHAR(255), PRIMARY KEY (doctor_id));</pre>

Appointments	4 REATE TABLE appointments (appointment_id INT(11) NOT NULL AUTO_INCREMENT, patient_name VARCHAR(100) NOT NULL, email VARCHAR(100) NOT NULL, patient_id INT(11), phone VARCHAR(15) NOT NULL, doctor_name VARCHAR(100) NOT NULL, appointment_date DATE NOT NULL, appointment_time TIME NOT NULL, message TEXT, submitted_at TIMESTAMP 7 NOT NULL DEFAULT CURRENT_TIMESTAMP, status VARCHAR(20) NOT NULL DEFAULT 'pending', PRIMARY KEY (appointment_id), FOREIGN KEY (patient_id) REFERENCES patients(patient_id) ON DELETE SET NULL ON UPDATE CASCADE);
9 Contact_Messages	CREATE TABLE contact_messages (id INT(11) NOT NULL AUTO_INCREMENT, name VARCHAR(100), email VARCHAR(100), subject VARCHAR(150), message TEXT, submitted_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP, PRIMARY KEY (id));
10 Doctor_schedule	CREATE TABLE doctor_schedule (schedule_id INT(11) NOT NULL AUTO_INCREMENT, doctor_id INT(11), day_of_week VARCHAR(20), start_time TIME, end_time TIME,

	⁸ PRIMARY KEY (schedule_id), FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id) ON DELETE CASCADE ON UPDATE CASCADE);
Prescriptions	CREATE TABLE prescriptions (prescription_id INT(11) NOT NULL AUTO_INCREMENT, ¹⁷ patient_id INT(11), doctor_id INT(11), date_issued DATE, medicines TEXT, notes TEXT, PRIMARY KEY (prescription_id), ¹ FOREIGN KEY (patient_id) REFERENCES patients(patient_id) ON DELETE SET NULL ON UPDATE CASCADE, FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id) ON DELETE CASCADE ON UPDATE CASCADE);
Test_report	CREATE TABLE test_reports (report_id INT(11) NOT NULL AUTO_INCREMENT, patient_id INT(11), doctor_id INT(11), report_type VARCHAR(100), file_path VARCHAR(255), uploaded_on DATETIME ⁵ NOT NULL DEFAULT CURRENT_TIMESTAMP, comments TEXT, PRIMARY KEY (report_id), ¹ FOREIGN KEY (patient_id) REFERENCES patients(patient_id) ON DELETE SET NULL ON UPDATE CASCADE, FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id) ON DELETE CASCADE ON UPDATE CASCADE);
Messages	CREATE TABLE messages (message_id INT(11) NOT NULL AUTO_INCREMENT, sender_role ENUM('admin','doctor','patient'),

	<pre>sender_email VARCHAR(150), sender_id INT(11), receiver_role ENUM('admin','doctor','patient'), receiver_email VARCHAR(150), receiver_id INT(11), content TEXT, sent_on DATETIME DEFAULT CURRENT_TIMESTAMP, PRIMARY KEY (message_id));</pre>
Payments	<pre>CREATE TABLE payments (payment_id INT(11) NOT NULL AUTO_INCREMENT, patient_id INT(11), appointment_id INT(11), amount DECIMAL(10,2), status VARCHAR(30) DEFAULT 'pending', paid_on DATETIME, PRIMARY KEY (payment_id), FOREIGN KEY (patient_id) REFERENCES patients(patient_id) ON DELETE SET NULL ON UPDATE CASCADE, FOREIGN KEY (appointment_id) REFERENCES appointments(appointment_id) ON DELETE SET NULL ON UPDATE CASCADE);</pre>
Feedback	<pre>CREATE TABLE feedback (feedback_id INT(11) NOT NULL AUTO_INCREMENT, patient_id INT(11) NOT NULL, doctor_id INT(11) NOT NULL, rating INT(11) NOT NULL, comment TEXT,</pre>

	6 created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, PRIMARY KEY (feedback_id), FOREIGN KEY (patient_id) REFERENCES patients(patient_id) ON DELETE CASCADE ON UPDATE CASCADE, FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id) ON DELETE CASCADE ON UPDATE CASCADE);
--	---

Table 6 – Queries

Task 03

Software testing ensures the system functions correctly, securely, and efficiently before deployment. For the MediCare Plus System, testing ensures that doctors, patients, and administrators experience a seamless workflow without errors or data loss.

a) Testing Overview

Testing phases:

- Unit and Integration Testing (backend functions)
- Functional testing (end-to-end flow login, signup, appointment, and file upload)
- Usability testing (users from sample demographic)
- Comparability testing (desktop/mobile /tablet)
- Performance testing (page load, DB query times)
- Security testing (SQL and session handling)
- Acceptance testing

b) Testing Plan

Test Objectives

- Verify workflows work end-to-end and boundary cases are handled.
- Validate security controls and data integrity.
- Ensure UI is usable and responsive.
- Checks error handling, input validations, and redirects.

Test Environment

- Local: XAMPP
- Brower: Chrome latest, Edge

Scope

In scope;

- | | |
|-----------------------|----------------------|
| • Functional testing | • Security testing |
| • UI validation | • System integration |
| • Database validation | • Error handling |

Out of Scope;

- Performance stress testing

Testing Types Performed

Functional Testing

- Patient signup and login
- Doctor login
- Booking an appointment
- Viewing appointments (Doctor+Admin+Patient)
- Uploading and viewing test reports
- Messaging between doctor and patient.
- Admin: Managing doctors, appointments, and reports.

Non-Functional Testing

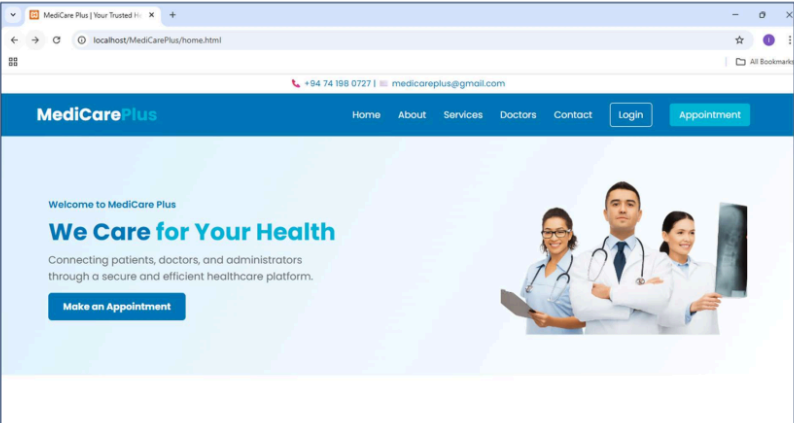
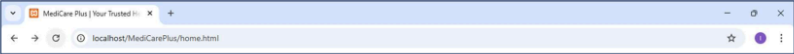
- 1) Performance Testing
 - Page load time under 1.5 seconds
 - Tested with 100+ appointment records
 - Dashboard loads fast without delay
- 2) Usability Testing
 - Navigation sidebar clear and consistent
 - Buttons and forms are easy to use
 - Responsive layout with readable typography
- 3) Security Testing
 - Password hashing
 - SQL injection was prevented using prepared statements
 - Session validation for role-based access
 - File uploads restricted to PDF only
- 4) Compatibility Testing
 - Works on Chrome, Firefox, Edge
 - Tested on desktop + laptop
 - No layout breakage

Test Cases

- **Test Case 1**

ID	TC01
Case	Homepage Load
Steps	Open the website with the URL in the browser. <i>http://localhost/MediCarePlus/home.html</i>
Expected Result	Homepage loads without errors.
Actual Result	Works
Result	Pass

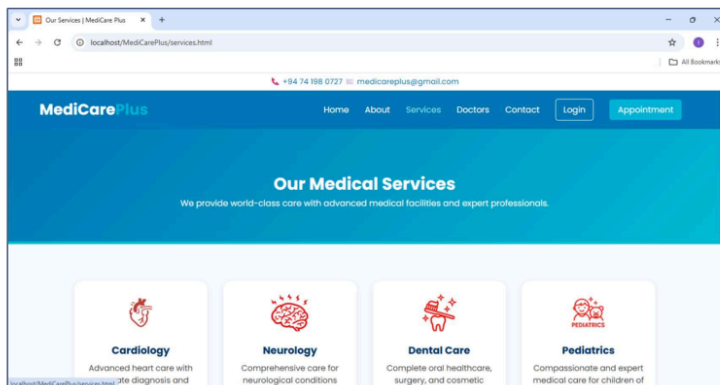
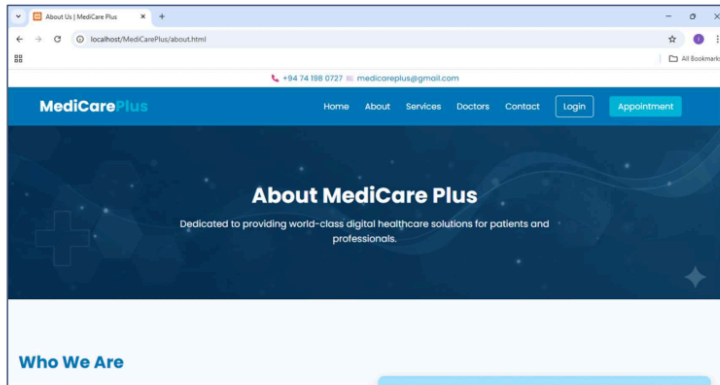
Evidence



- 3 **Test Case 2**

ID	TC02
Case	Navigation Links
Steps	Click all navigation links
Expected Result	Each link redirects to the correct page
Actual Result	Works
Result	Pass

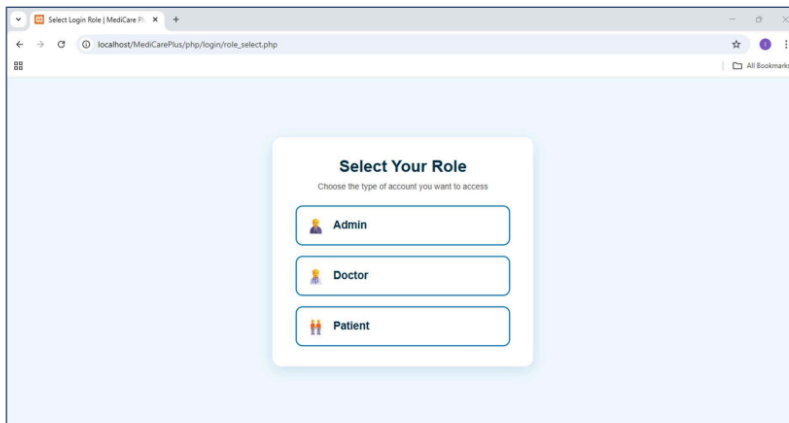
Evidence



- **Test Case 3**

ID	TC03
Case	Role Selection Page
Steps	Open login_role.html Click each role
Expected Result	Goes to correct login pages
Actual Result	Works
Result	Pass

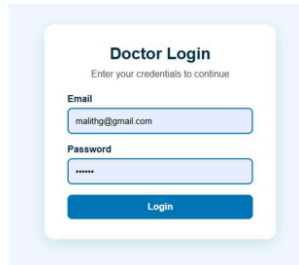
Evidence



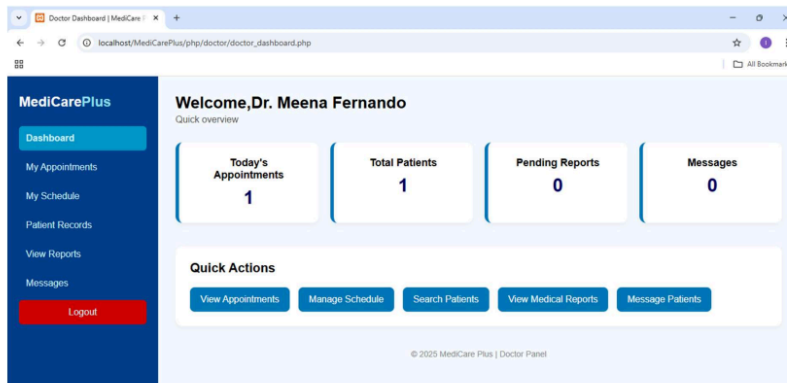
- Test Case 4

ID	TC04
Case	Doctor Login
Steps	Enter a valid email and password
Expected Result	Goes to correct dashboard pages
Actual	Works
Result	Pass

Evidence:



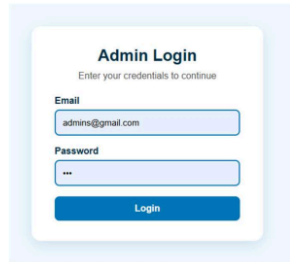
The screenshot shows a 'Doctor Login' form with the subtitle 'Enter your credentials to continue'. It features two input fields: 'Email' with the value 'malith@gmail.com' and 'Password' with masked characters. A blue 'Login' button is positioned below the password field.



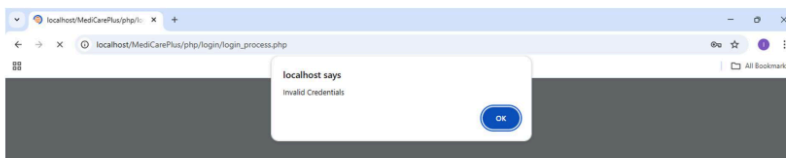
- **Test Case 5**

ID	TC05
Case	Invalid login
Steps	Enter the wrong credentials
Expected Result	Show “invalid login” messages
Actual	Works
Result	Pass

Evidence



The image shows a web form titled "Admin Login". Below the title is a subtitle "Enter your credentials to continue". There are two input fields: "Email" with the value "admins@gmail.com" and "Password" with a masked value "***". A blue "Login" button is at the bottom.



- **Test Case 6**

ID	TC03
Case	Appointment booking
Steps	The patient books an appointment with a valid date and time, specifying a particular doctor.
Expected Result	Record inserted into the appointments database Redirect Successful
Actual	Record inserted Appointments table updated
Result	Pass

Evidence

Book an Appointment

Please fill out the form below to schedule your visit.

Tehan Perera

tehan@gmail.com

0742174362

Dr. Dilin Fernando - Dentist

11/26/2025

07:00 AM

Consultation for teeth pain

Book Appointment

Book an Appointment

Please fill out the form below to schedule your visit.

Appointment booked successfully!

Browser: 127.0.0.1 - localhost:3000 - idm-appointments

Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing 13 rows (13 total) | 13 rows | 13 rows

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	appointment	patient_name	email	phone	doctor_name	appointment_date	appointment_time	message	submitted_at
☐	Edit Copy Delete	4	Sarah Perera	Sarah@gmail.com	074155625	2025-11-08	23:58:00	qqq	2025-11-09 20:55:12
☐	Edit Copy Delete	5	Sarah Perera	Sarah@gmail.com	011217895	2025-11-12	02:50:00	I need to know about the report results.	2025-11-09 00:48:13
☐	Edit Copy Delete	6	Sarah Perera	vihara@gmail.com	222222	2025-11-16	15:43:00	ssssssss	2025-11-14 22:43:52
☐	Edit Copy Delete	7	Sarah Perera	Sarah@gmail.com	074155625	2025-11-22	10:30:00	Routine checkup	2025-11-20 16:41:00
☐	Edit Copy Delete	8	Vihara Sandamal	vihara@gmail.com	077123456	2025-11-28	14:00:00	I need medicine refil	2025-11-21 16:30:00
☐	Edit Copy Delete	9	Kasun Perera	kasun@gmail.com	071223333	2025-11-30	09:45:00	Dental checkup	2025-11-21 18:22:00
☐	Edit Copy Delete	10	Nimal Perera	malith@gmail.com	Nimal Perera	2025-11-26	22:00:00		2025-11-22 18:59:06
☐	Edit Copy Delete	11	Malith Gunasekara	malith@gmail.com	0112145627	2025-11-23	22:02:00	I will be coming 15 mins late	2025-11-22 19:01:50
☐	Edit Copy Delete	12	Malith Gunasekara	malith@gmail.com	0112121	2025-11-24	19:14:00	Want to check my reports and get medicine.	2025-11-22 23:10:02
☐	Edit Copy Delete	13	Tehan Perera	tehan@gmail.com	0742174362	2025-11-26	07:00:00	Consultation for teeth pain	2025-11-23 01:31:00

- Test Case 7

ID	TC07
Case	Duplicate Email Signup
Steps	Submit with an already used email
Expected Result	Error: Email already registered.
Actual	Error shows
Result	Pass

Evidence

Create Patient Account

Fill in the details to register

Full Name

Email

Phone

Password

[Create Account](#)

Already have an account? [Login](#)

Create Patient Account

Fill in the details to register

Email already registered

Full Name

Email

Phone

Password

[Create Account](#)

Already have an account? [Login](#)

- **Test Case 8**

ID	TC08
Case	View Doctor Appointments
Steps	The patient checks the appointments page
Expected Result	Table loads all assigned appointments
Actual	Shows all appointments to the relevant doctor.
Result	Pass

Test Data

- Email: meena@medicare.com
- Password: malith

Evidence

My Appointments | MediCare

localhost/MediCarePlus/phys/patient/appointments_patient.php

GmailYouTubeMapsNew folderPin pageHome | Cardiff Met...Select Login Role |...MediCare Plus | You...All Bookmarks

MediCarePlus

Dashboard

My Appointments

Prescriptions

Test Reports

Payments

Messages

Login

My Appointments

Monday, November 24, 2025

+ Book New Appointment

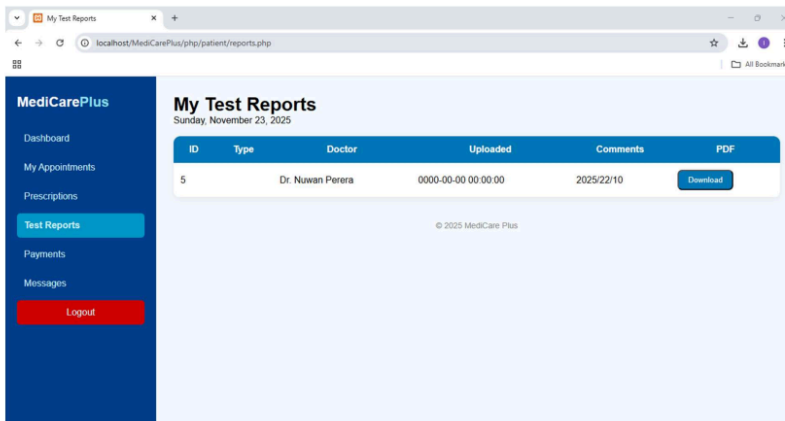
ID	Doctor	Date	Time	Phone	Message	PDF
10	Dr. Anusha de Silva	2025-11-26	10:00 PM	Nimal Perera		Download
12	Dr. Anusha de Silva	2025-11-24	07:14 PM	011213	Want to check my reports and get medicine.	Download
11	Dr. Anusha de Silva	2025-11-23	10:02 PM	01121459827	I will be coming 15 mins late	Download
15	Dr. Asanka Senanayake	2025-11-06	04:24 PM	011213sss	sssss	Download

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- **Test Case 9**

ID	TC09
Case	View Test Reports
Steps	Patient/Doctor checks reports
Expected Result	PDF downloads
Actual	Can see available test reports Can download pdf
Result	Pass

Evidence



- **Test Case 10**

ID	TC10
Case	Send Message
Steps	Patient sends a message to the doctor
Expected Result	Message stored Visible to the doctor
Actual	In the doctor dashboard, messages are visible of the relevant patient's email.
Result	Pass

Evidence

Messages

Send to: Doctor

Receiver email meena@medicare.com

Message Did you check my reports

Send

Messages

Send to: Doctor

Receiver email

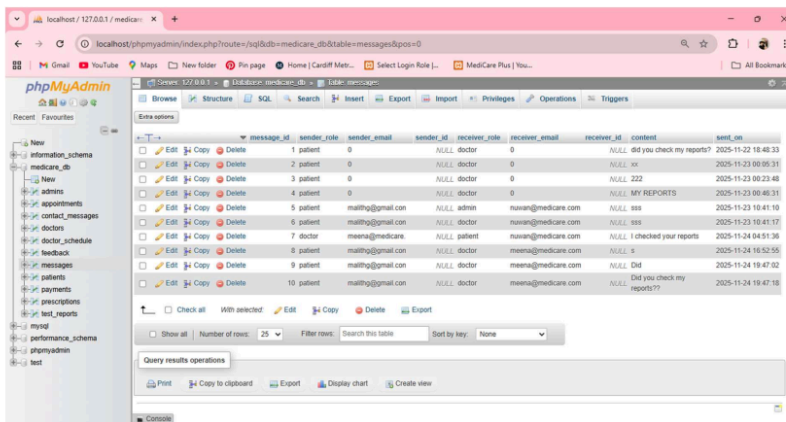
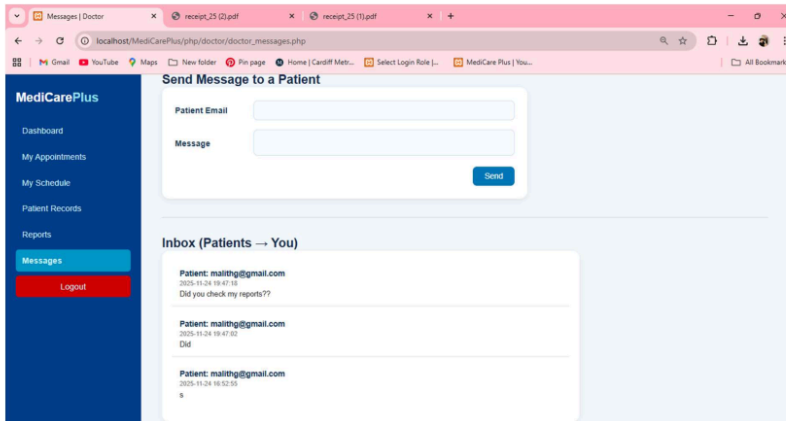
Message

Send

Recent Messages

patient.malith@gmail.com → meena@medicare.com
Did you check my reports??

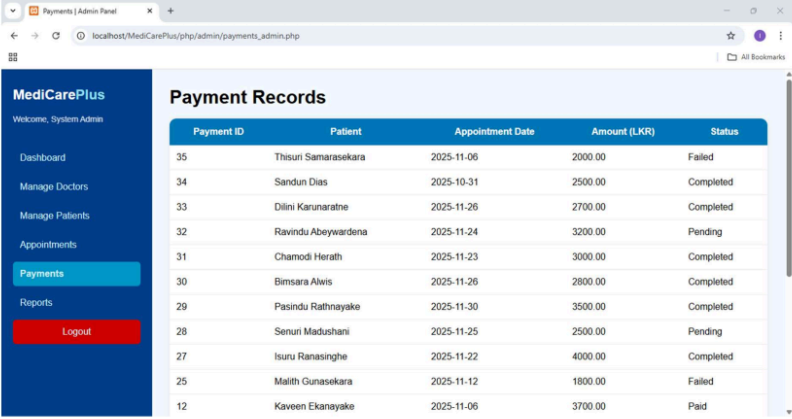
2025-11-24 19:47:18



- Test Case 11

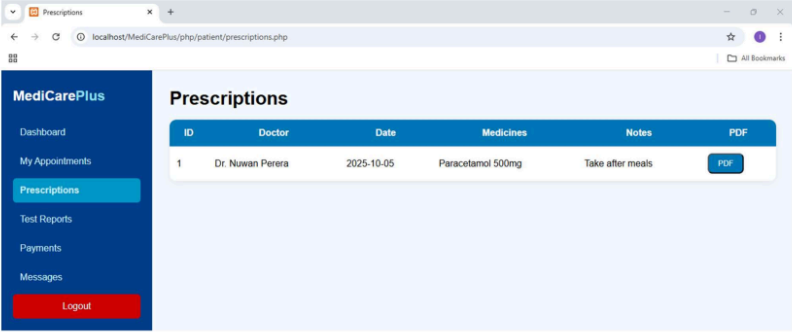
ID	TC11
Case	Sidebar activate state
Steps	Checking the sidebar of the dashboards.
Expected Result	Current page link highlighted.
Actual	Current link is highlighted
Result	Pass

Evidence



The screenshot shows the 'Payments | Admin Panel' interface. The sidebar on the left contains the following menu items: Dashboard, Manage Doctors, Manage Patients, Appointments, **Payments**, Reports, and a Logout button. The main content area is titled 'Payment Records' and contains a table with the following data:

Payment ID	Patient	Appointment Date	Amount (LKR)	Status
35	Thisun Samarasekara	2025-11-06	2000.00	Failed
34	Sandun Dias	2025-10-31	2500.00	Completed
33	Dilini Karunaratne	2025-11-26	2700.00	Completed
32	Ravindu Abeywardena	2025-11-24	3200.00	Pending
31	Chamodi Herath	2025-11-23	3000.00	Completed
30	Bimsara Alwis	2025-11-26	2800.00	Completed
29	Pasindu Rathnayake	2025-11-30	3500.00	Completed
28	Senuri Madushani	2025-11-25	2500.00	Pending
27	Isuru Ranasinghe	2025-11-22	4000.00	Completed
25	Malith Gunasekara	2025-11-12	1800.00	Failed
12	Kaveen Ekanayake	2025-11-06	3700.00	Paid



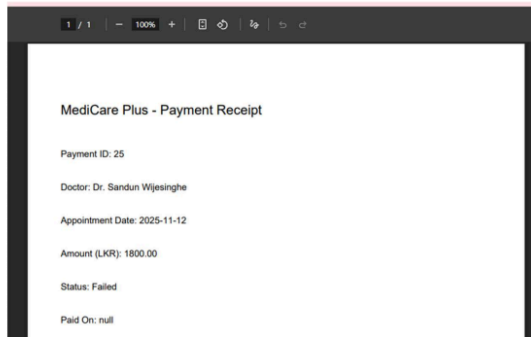
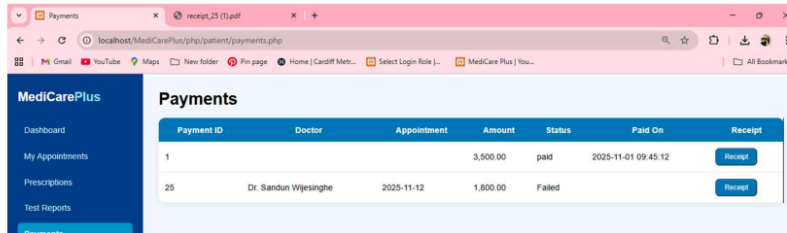
The screenshot shows the 'Prescriptions' interface for a patient. The sidebar on the left contains the following menu items: Dashboard, My Appointments, **Prescriptions**, Test Reports, Payments, Messages, and a Logout button. The main content area is titled 'Prescriptions' and contains a table with the following data:

ID	Doctor	Date	Medicines	Notes	PDF
1	Dr. Nuwan Perera	2025-10-05	Paracetamol 500mg	Take after meals	PDF

- **Test Case 12**

ID	TC12
Case	Payment Receipt PDF
Steps	Patient dashboard Payment side downloading pdf
Expected Result	Can download the pdf successfully
Actual	PDF downloaded
Result	Pass

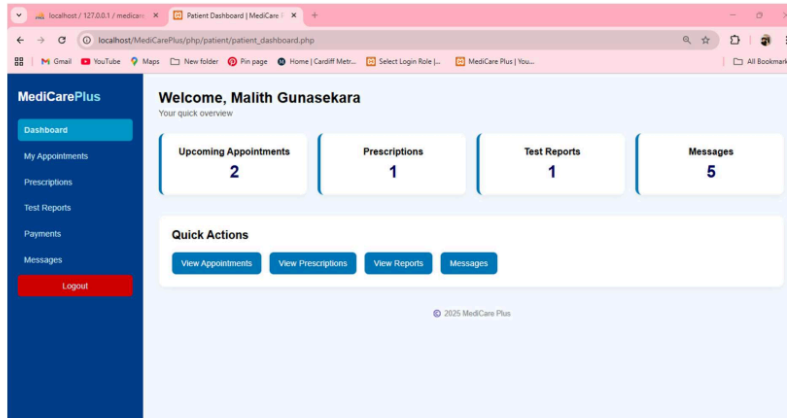
Evidence



- **Test Case 14**

ID	TC14
Case	Page load time
Steps	Load dashboard
Expected Result	Load dashboard within 2 secs
Actual	Loaded patient dashboard within 2 secs.
Result	Pass

Evidence



- Test Case 15

ID	TC15
Case	Add Doctor
Steps	In the admin dashboard, in Manage Doctors, add a doctor.
Expected Result	Data appears in a <i>table</i>
Actual Result	Data stored
Result	Pass

Evidence

Add New Doctor

Doctor Name

Specialization

Email

Phone

Password

Add Doctor

Manage Doctors

All Registered Doctors

+ Add Doctor

ID	Name	Specialization	Email	Phone	Actions
47	Dr.Hematha Perera	Dentist	hemantha@gmail.com	0775981569	Delete

5) Feedback Evolution

Feedback was collected from two students during the demonstration phase.

User Feedback Summary

Feedback Source	Comments	Modifications
Student 1	Improve navigation clarity	Added colored hover effects and icons
Student 2	The appointment form is too long	Grouped fields and improved spacing.

After incorporating this feedback;

- UI improved significantly.
- User flow became smoother.
- Dashboard became clearer.

6) Testing Conclusion

The MediCare Plus web application successfully met all the testing objectives. All critical features including login, registration, appointments, dashboards, CRUD operations, and navigation performed as expected. Major functional bugs remain, and the system is ready for deployment.

The testing phase validated:

- Correctness and efficiency of backend functionality
- Responsiveness and usability of UI designs
- Security and session handling across dashboards
- Database integrity with valid, consistent data

The application meets the academic requirements for a high-quality, user-friendly medical web system.

Conclusion

MediCare Plus is a groundbreaking web-based healthcare platform thoughtfully crafted to revolutionize the experience for patients and healthcare providers alike. Our initiative primarily focuses on architecting an intuitive, user-friendly interface that not only simplifies healthcare management but also promotes engagement and empowerment among users.

Key features of MediCare Plus include:

- **Effortless Appointment Scheduling:** Users can seamlessly book, reschedule, or cancel appointments with just a few clicks. The platform's intuitive calendar interface displays real-time availability, significantly reducing wait times and ensuring accessibility for every user.
- **Comprehensive Telehealth Capabilities:** MediCare Plus supports robust virtual consultations, allowing healthcare providers to interact with patients through secure video conferencing. This feature ¹⁴is particularly beneficial for individuals residing in rural areas or those facing mobility limitations, clearly expanding their access to quality healthcare.
- **Secure Messaging System:** Our platform facilitates confidential and efficient communication between patients and providers through a secure messaging system. This feature enables users to pose quick inquiries, seek follow-up information, and clarify details—all from the comfort of their homes, without necessitating in-person visits.
- **Integrated Health Records System:** MediCare Plus boasts a sophisticated digital health records feature that compiles ¹¹a patient's medical history, lab results, prescriptions, and treatment plans into one cohesive platform. This comprehensive integration enhances ¹¹healthcare providers' ability to make informed decisions and facilitates superior care coordination.

Ultimately, MediCare Plus aspires to enhance healthcare accessibility and streamline operational efficiency while keeping user needs at the forefront. By empowering individuals to take charge of their health and simplifying the workflows for providers, we aim to foster a more effective, responsive, and patient-centered healthcare system.

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